ADITYA UNIVERSITY

PROBABILITY AND STATISTICS

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Unit-3
Data Analytics and Sampling Theory

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Data Analytics:

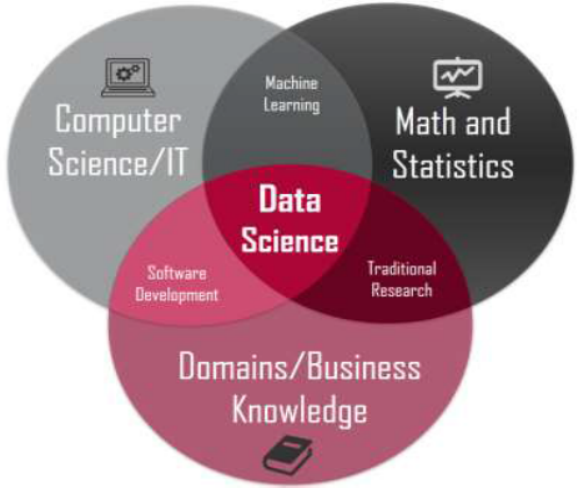
Data Science-Statistics Introduction-Population Vs Sample-Collection of data-Primary and secondary Data-Type of variable-:dependent and independent, Categorical and Continuous variables-Data visualization-Measures of Central tendency-Measures of variability(Spread or variance)

Data Science

Data science is an inter-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from structured and unstructured data, and apply knowledge and actionable insights from data across a broad range of application domains. Data science is related to data mining, machine learning and big data.

- a "concept to unify statistics, data analysis, machine learning and their related methods" in order to "understand and analyze actual phenomena" with data.

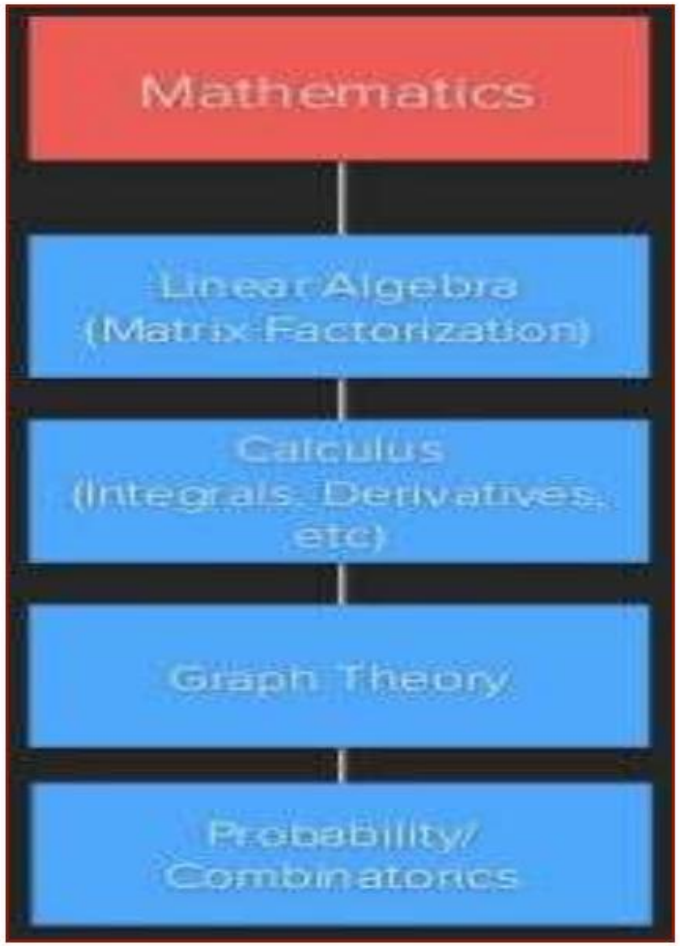
- employs techniques and theories drawn from many fields within the context of mathematics, statistics, computer science, and information science.

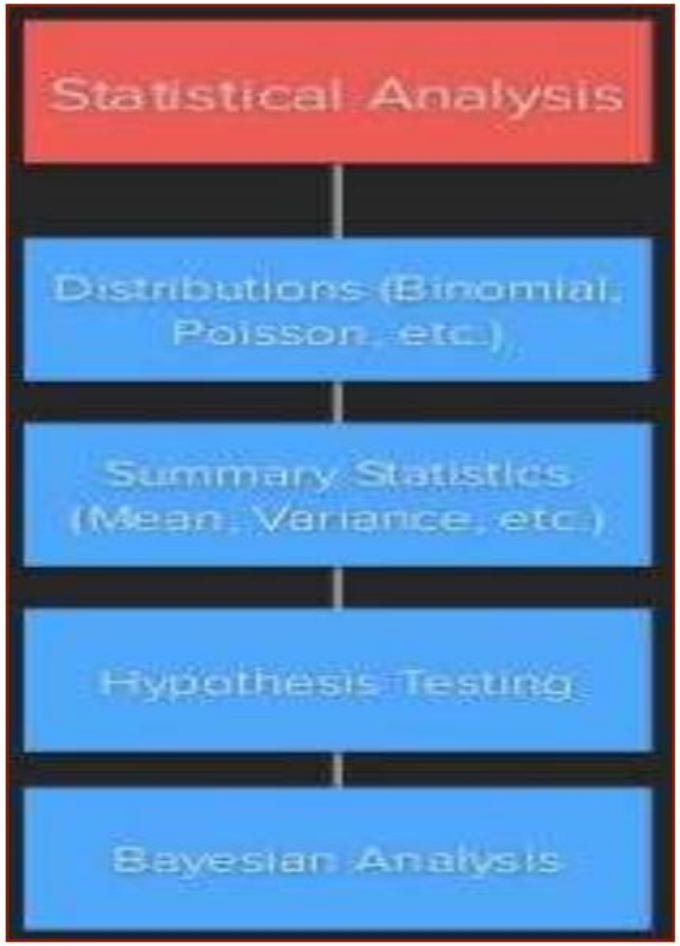


Regular Data Science

- Data Analysis
- Modelling Statistics
- Engineering / Prototyping

Data Scientist need to comfortable with:





- **Applications of Data Science**

- Security
- Sports
- Banking and Finance
- Internet Search
- Digital Advertisements
- Recommender System
- Image Processing
- Speech Recognition
- Gaming
- Price Comparison Websites
- Airline Routing Planning
- Fraud and Risk Detection
- Delivery Logistics
- Internet of Things (IoT)
- Health Care
- Augmented Reality
- Self-Driving Cars
- Robots

Importance of Data Science

1. Data science helps brands to understand their customers in a much enhanced and empowered manner.
2. It allows brands to communicate their story in such a engaging and powerful manner.
3. Big Data is a new field that is constantly growing and evolving.

4. Its findings and results can be applied to almost any sector like travel, healthcare and education among others.

5. Data science is accessible to almost all sectors.

STATISTICS

Statistics is the discipline that concerns the collection, organization, analysis, interpretation, and presentation of data.

Statistics deals with every aspect of data, including the planning of data collection in terms of the design of surveys and experiments

When census data cannot be collected, statisticians collect data by developing specific experiment designs and survey samples

- Descriptive statistics
- Inferential statistics.

- Organizing and summarizing data using informal methods like graphing and using numbers is called descriptive statistics.

- Organizing and summarizing data using formal methods is called Inferential statistics.

ROLE OF STATISTICS

- Statistics is a language of data
- Statistics provides a scientific way to extract and retrieve the information hidden inside the data
- Statistics cannot do miracles
- Statistics cannot change the process or phenomenon
- Statistical tools provide forecasting buy not like astrologer's parrot
- Statistics cannot do anything in no time .Tools can be developed but development needs time

STATISTICAL TOOLS

Several tools and components are available

- Graphical tools
  provide visualization from first hand information
- Analytical Tools
  provide quantitative information

Both approaches work together and are inseparable

STATISTICAL TOOLS

Graphical Tools	Analytical Tools
2D&3D plots	Central tendency of data: mean, median ,mode, geometric mean, harmonic mean, quartiles
Scatter Diagram	
Pie Diagram	Dispersion of data: Variance, standard deviation, standard error, mean deviation, absolute deviation, range etc.,
Histogram	
Bar Chart	
Stem & Leaf plot	
Box Plot	

Population Vs Sample

Population(or Universe): A collection or aggregate or totality of persons, things or objects or statistical data under study is known as population.

Size of the Population: Number of units in the population is known as size of the population. It is denoted by N.

Finite Population: The number of units in the population is finite then it is known as finite population.

Infinite Population: : The number of units in the population is infinite then it is known as infinite population.

Sample: A portion(or subset) of the population is known as sample.
Size of the sample: Number of units in the sample is known as size of the sample. It is denoted by n.
Large sample: If the size of the sample is greater than 30 then it is known as large sample.
Small sample: If the size of the sample is less than 30 then it is known as small sample.
Parameter: The values that are obtained from the population data or a number that describes the property of the population is known as parameters
Statistic: The values that are obtained from the sample data or a number that describes the property of the sample is known as Statistics.

Collection of Data

Types of data:

Primary data

Secondary data

Primary Data:

Data originally collected by an investigator for the first time for any statistical investigation and it is the type of **data** that is collected by researchers directly from main **sources.**

Sources of primary data:

- Direct personal investigation
- Questionnaire received through postal mail, e-mail, e-forms(google forms), online surveys etc.,
- Questionnaire sent through surveyors

Secondary data:

Data which has already been collected by some person or agency for any statistical investigation

Sources of secondary data:

- Published sources ex: books, journals, articles, web pages etc.
- Data collected from survey agencies
- public reports the data Ex: municipalities
- blogs etc
- library, bots, internet sources etc.

Examples:

An example of primary data is the national census data collected by the government while an example of secondary data is the data collected from online sources.

The secondary data collected from an online source could be the primary data collected by another researcher.

For example, the government, after successfully the national census, they share the results in newspapers, online magazines, press releases, etc. Another government agency that is trying to allocate the state budget for healthcare, education, etc. may need to access the census results.

With access to this information, the number of children who needs education can be analyzed and hard to determine the amount that should be allocated to the education sector. Similarly, knowing the number of old people will help in allocating funds for them in the health sector.

Advantages:

Some common advantages of primary data are its authenticity, specific nature, and up to date information while secondary data is very cheap and not time-consuming.

Primary data is very reliable because it is usually objective and collected directly from the original source. It also gives up to date information about a research topic compared to secondary data.

Secondary day, on the other hand, is not expensive making it easy for people to conduct secondary research. It doesn't take so much time and most of the secondary data sources can be accessed for free.

Similarities Between Primary & Secondary Data

1. Contains Same Content:

Secondary data was once primary data when it was newly collected by the first researcher. The content of the data collected does not change and therefore has the same content with primary data.

It doesn't matter if it was further visualized in the secondary form, the content does not change. A common example of these are definitions, theorems, and postulates that were made years ago but still remain the same.

2.Primary data and secondary data both have applications in business and research. They may, however, differ from each other in the way in which they are collected, used, and analyzed.

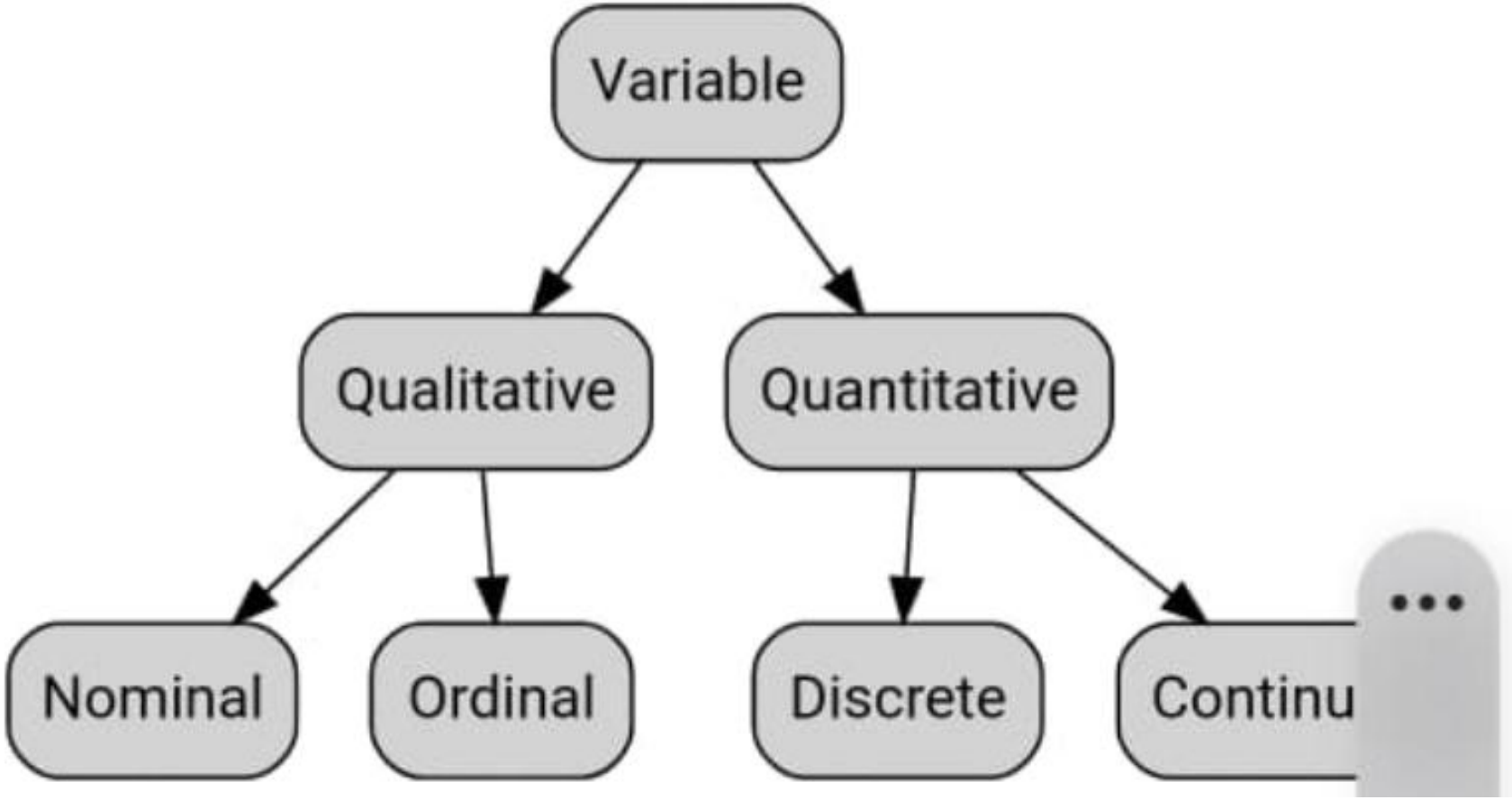
Differences between primary and secondary data:

1. Primary data is very expensive while secondary data is economical. When working on a low budget, it is better for researchers to work with secondary data, then analyze it to uncover new trends.
2. Primary data is more accurate and reliable while secondary data is relatively less reliable and accurate. This is mainly because the secondary data sources are not regulated and are subject to personal bias.
3. Primary data is available in crude form while secondary data is available in a refined form. That is, secondary data is usually made available to the public in a simple form for a layman to understand while primary data are usually raw and will have to be simplified by the researcher.

Variables and types of variables

The values that are altering according to circumstances are referred to as variables. A variable can occurs in any form, such as trait, factor or a statement that will constantly be changing according to the changes in the applied environment.

In statistics, variables are broadly divided into two main categories: **Categorical (Qualitative)** and **Numerical (Quantitative)**. Categorical variables represent groups or labels, while numerical variables represent measurable amounts.


Variable
Qualitative
Quantitative
Nominal
Ordinal
Discrete
Continu
...

1.Categorical (Qualitative) Variables

These variables represent characteristics, descriptions, or labels that cannot be quantified mathematically.

Nominal: Variables with named categories that have **no intrinsic order**.

Examples: Eye color (blue, brown, green), brands of smartphones, gender.

Ordinal: Variables with categories that **do have a natural, meaningful order** or ranking.

Examples: Customer satisfaction ratings (poor, good, excellent), educational levels (high school, bachelor's, master's).

2. Numerical (Quantitative) Variables

These variables represent measurable amounts and can be expressed numerically.

Discrete: Variables that can only take specific, countable, or whole number values.

Examples: Number of cars in a parking lot, number of children in a family, roll of a die.

Continuous: Variables that can take an infinite number of values within a given range, including decimals and fractions.

Examples: Height (5.9 feet), weight (72.5 kg), time taken to complete a task.

3. Variables in Experimental Design

When analyzing cause-and-effect relationships, statisticians use these specific roles.

Independent Variable (IV): The variable you manipulate or change to see its effect on the outcome.

Example: The amount of fertilizer given to a plant.

Dependent Variable (DV): The variable you measure or observe to see the outcome of the manipulation.

Example: The height of the plant after three weeks.

Controlled Variable: Factors kept constant to ensure the experiment is fair.

Example: The amount of sunlight and water the plant receives.

Data Visualization

Data visualization is the graphical representation of information and **data**.

The main goal of data visualization is to make it easier to identify patterns, trends and outliers in large data sets.

By using visual elements like charts, graphs, and maps, **data visualization** tools provide an accessible way to see and understand trends, outliers, and patterns in **data**. In the world of Big Data, data visualization tools and technologies are essential to analyze massive amounts of information and make data-driven decisions.

Benefits of data visualization :

•the ability to absorb information quickly, improve insights and make faster decisions

•an increased understanding of the next steps that must be taken to improve the organization

•an improved ability to maintain the audience's interest with information they can understand

•an easy distribution of information that increases the opportunity to share insights with everyone involved

•eliminate the need for data scientists since data is more accessible and understandable and

•an increased ability to act on findings quickly and, therefore, achieve success with greater speed and less mistakes.

Common general types of data visualization:

- Charts
- Tables
- Graphs
- Maps
- Info graphics
- Dashboards

More specific examples of methods to visualize data:

- Bar Chart
- Pie chart
- Histogram
- Stem leaf plots
- Scatter diagrams etc.

Absolute frequency:

Number of times that a value appears is known as absolute frequency. It is represented as f_i where the subscript represents each of the values. The sum of the absolute frequencies is equal to the total number of data, represented bas N.

$$N = f_1 + f_2 + - - - + f_n = \sum_{i=1}^{n} f_i$$

Relative frequency: The result of dividing the absolute frequency of a certain value by the total number of data. It is represented as n_i . The sum of the relative frequencies is equal to 1. We can prove this easily by factorizing N.

$$n_i = \frac{f_i}{N}$$

Example

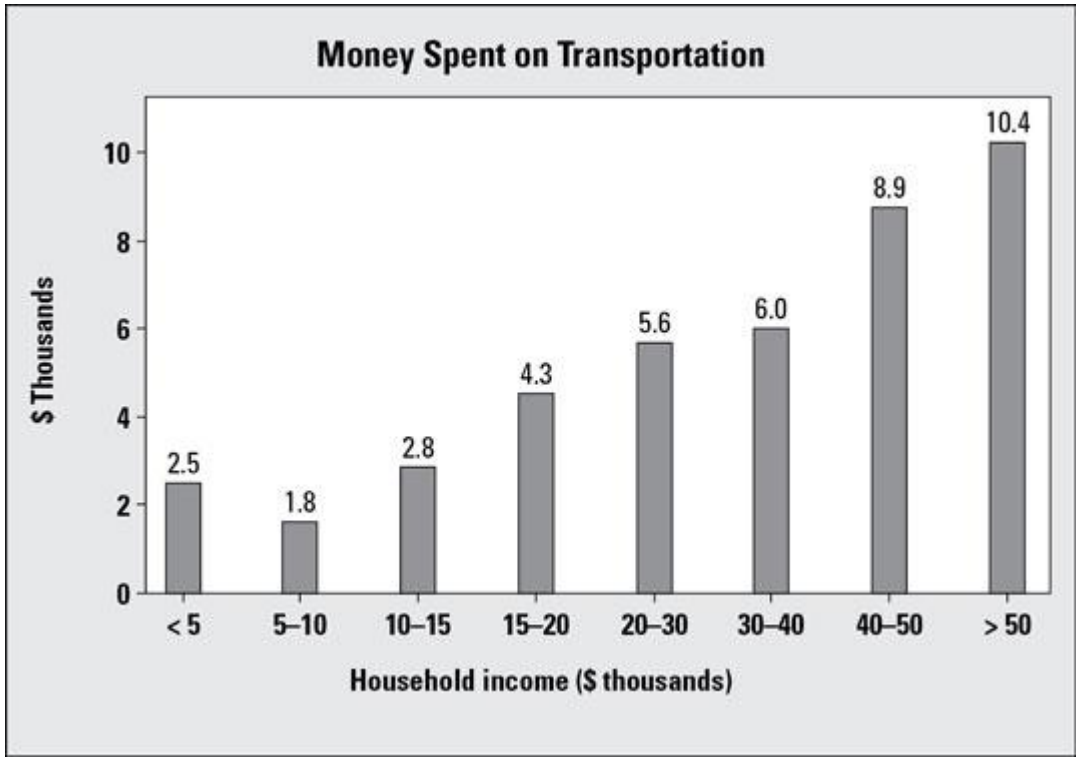
15 students answer the question of how many brothers or sisters they have. The answers are:
1,1,2,0,3,2,1,4,2,3,1,0,0,1,2
Then, we can construct a table of frequencies

Solution:
Here
N=3+5+4+2+1=15

Brothers	Absolute frequency fi	Relative frequency ni	Cumulative frequency Fi	Relative cumulative frequency Ni
0	3	3/15	3	3/15
1	5	5/15	3+5=8	3/15+5/15 =8/15
2	4	4/15	3+5+4=12	12/15
3	2	2/15	3+5+4+2= 14	14/15
4	1	1/15	3+5+4+2+ 1=15	1

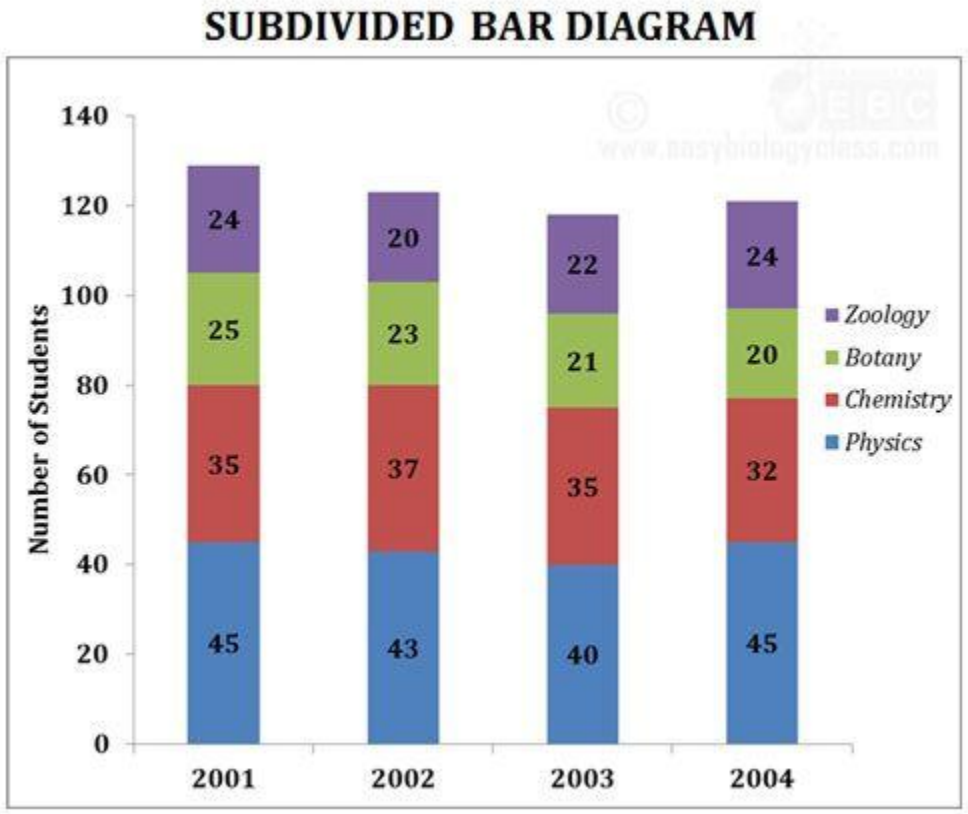
Bar diagrams:

- Visualizes the relative or absolute frequencies of observed values of a variable
- Consists of one bar for each category either horizontal or vertical
- Height of each bar is decided by the frequency of the respective category shown in y-axis



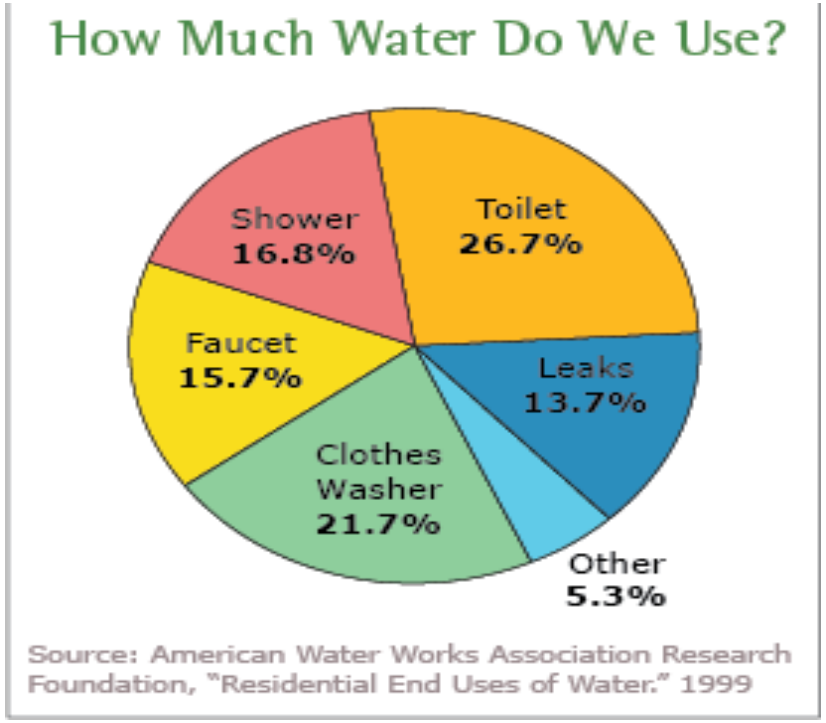
Subdivided Bar diagrams:

- Divides the total magnitude of variables into various parts



Pie diagrams:

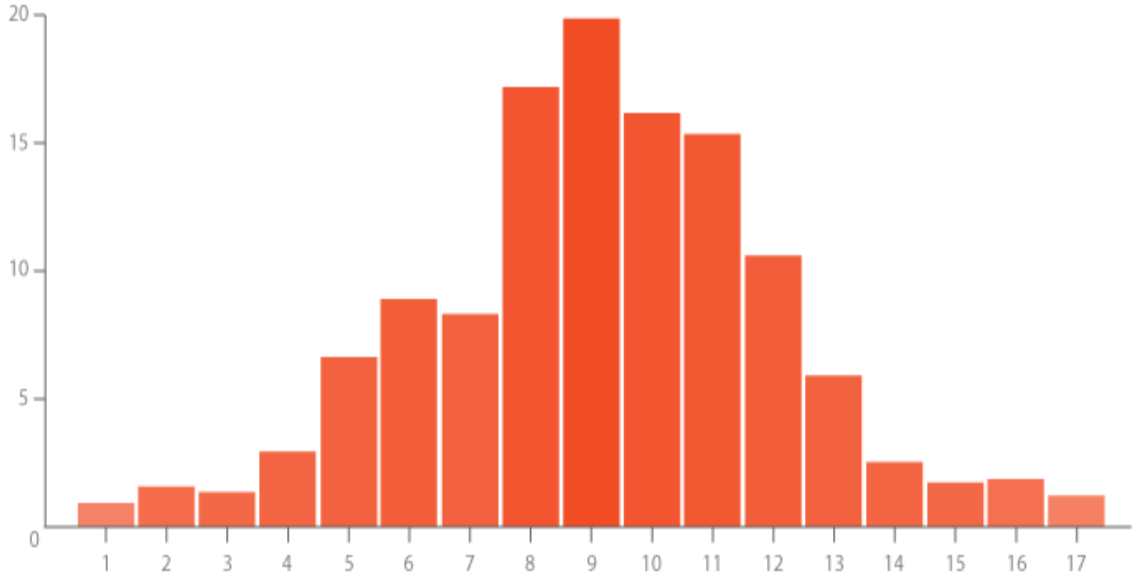
- Visualizes the absolute and relative frequency
- A circle partitioned into segments where each of the segments represents a category
- Size of each segment depends upon relative frequency and is determined by the angle(relative frequency x 360°)



Histogram:

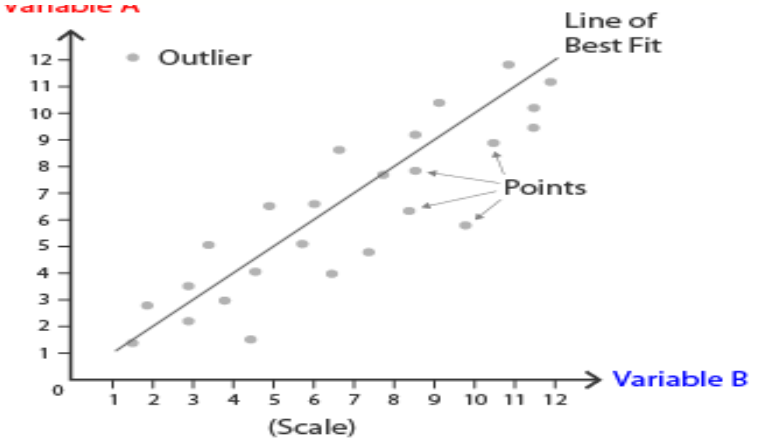
A Histogram visualizes the distribution of data over a continuous interval or certain time period. Each bar in a histogram represents the tabulated frequency at each interval/bin.

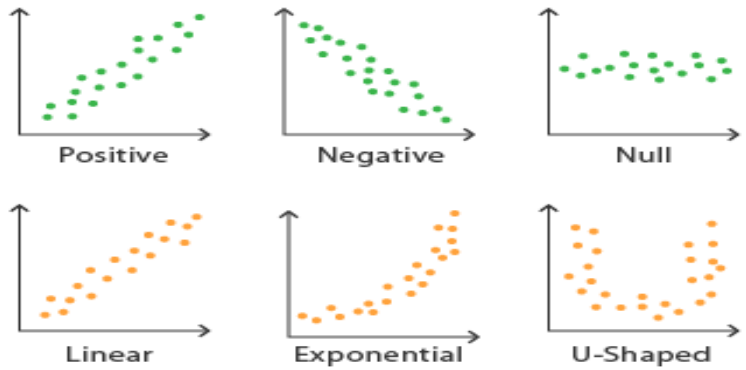
Histograms help give an estimate as to where values are concentrated, what the extremes are and whether there are any gaps or unusual values. They are also useful for giving a rough view of the probability distribution.



Scatter plot:

Also known as a *Scatter Graph, Point Graph, X-Y Plot, Scatter Chart or Scattergram*.
Scatterplots use a collection of points placed using Cartesian Coordinates to display values from two variables. By displaying a variable in each axis, you can detect if a relationship or correlation between the two variables exists.







Various types of correlation can be interpreted through the patterns displayed on Scatterplots. These are: **positive** (values increase together), **negative** (one value decreases as the other increases), **null** (no correlation), **linear**, **exponential** and **U-shaped**. The strength of the correlation can be determined by how closely packed the points are to each other on the graph. Points that end up far outside the general cluster of points are known as **outliers**.

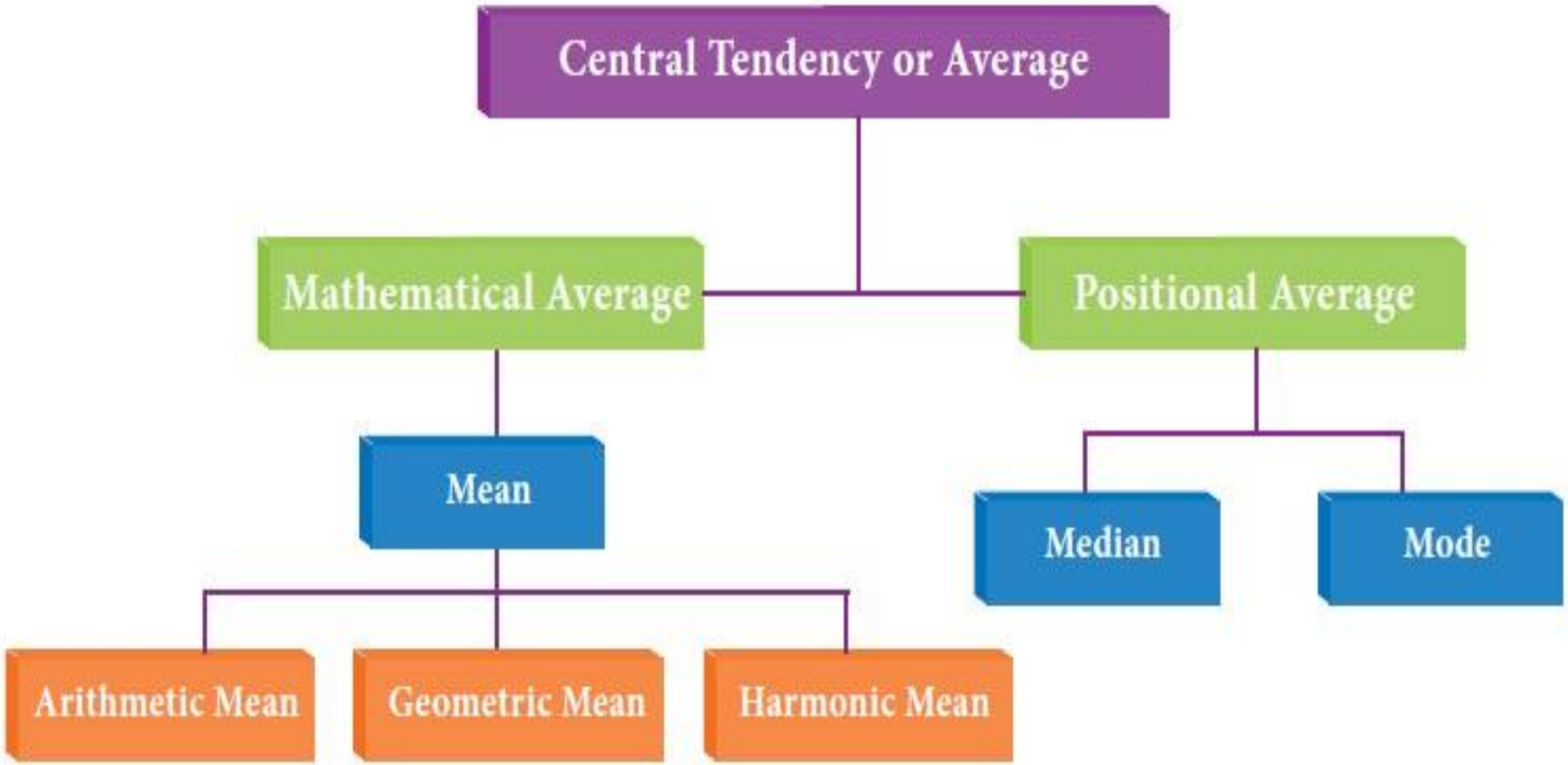
Measures of Central tendency:

A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data or the statistical measure that represents a single value of the entire dataset.

It is also called as measures of central location or summary statistics.

The five measures of central tendency are

1. Arithmetic mean or simple mean
2. Median
3. Mode
4. Geometric mean and
5. Harmonic mean


Central Tendency or Average
Mathematical Average
Positional Average
Mean
Median
Mode
Arithmetic Mean
Geometric Mean
Harmonic Mean

1. Arithmetic mean or simple mean:

It is the most popular and well known measure of central tendency. It can be used with both discrete and continuous data.

The arithmetic mean of a set of observations is the sum of observations divided by the number of observations in the data. It is denoted by $\bar{x}$

For n values in a set of data namely as x_1, x_2, x_3, ... x_n, the mean of data is given as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 \ldots\ldots\ldots x_n}{n}$$

or

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

For calculating the mean when the frequency of the observations is given, such that $x_1, x_2, x_3, \ldots, x_n$ is the recorded observations, and $f_1, f_2, f_3 \ldots f_n$ is the respective frequencies of the observations then;

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + f_3 x_3 \ldots\ldots\ldots f_n x_n}{f_1 + f_2 + f_3 \ldots\ldots\ldots f_n}$$

or

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

Example 1 :

The marks obtained by 10 students in a test are 15, 75, 33, 67, 76, 54, 39, 12, 78, 11. Find the arithmetic mean.

Solutior

Mean $\bar{x} = \frac{x_1 + x_2 + x_3 \dots \dots \dots \dots x_n}{n}$

=Total marks of 10 students / 10

= (15 + 75 + 33 + 67 + 76 + 54 + 39 + 12 + 78 + 11) / 10

= 460 / 10

Mean= 46

Example 2 :

John studies for 4 hours, 5 hours and 3 hours respectively on three consecutive days. How many hours does he study daily on an average?

Solution :

The average study time of John

= Total number of study hours / Number of days for which he studied

= (4 + 5 + 3) / 3

= 12 / 3

= 4 hours

Example 3

A batsman scored the following number of runs in six innings:

36, 35, 50, 46, 60, 55

Calculate the mean runs scored by him in an inning.

Solution :

To find the mean, we find the sum of all the observations and divide it by the number of observations.

Mean = Total runs / Number of innings

= (36 + 35 + 50 + 46 + 60 + 55) / 6

= 47

Thus, the mean runs scored in an inning are 47.

Arithmetic mean can be computed for both ungrouped data (raw data: data without any statistical treatment) and grouped data (data arranged in tabular form containing different groups).

Method's Name	Nature of Data	
	Ungrouped Data	Grouped Data
Direct Method	$\bar{x} = \frac{\sum x}{n}$	$\bar{x} = \frac{\sum fx}{N}$, $N = \sum f_i$
Indirect or Short-Cut Method	$\bar{x} = A + \frac{\sum D}{n}$, $D = x - A$	$\bar{x} = A + \frac{\sum fD}{N}$, $D = x - A$
Method of Step-Deviation	$\bar{x} = A + \frac{\sum u}{n}.c$, $u = (x - A)/c$ or h	$\bar{x} = A + \frac{\sum fu}{N}.c$

Provide the given distribution of the following frequency distribution of first year students of a particular college:

Age (Years)	13	14	15	16	17
Number of Students	2	5	13	7	3

Solution:

The given distribution is grouped data and the variable involved is ages of first year students, while the number of students represents frequencies.

Ages (Years) x	**Number of Students** f	fx
13	2	26
14	5	70
15	13	195
16	7	112
17	3	51
Total	∑f=30	∑fx=454

$$\bar{x} = \frac{\sum fx}{N}$$

$$N = \sum f_i$$

=454/30=15.13 years

Example:

The following data shows the distance covered by 100 people to perform their routine jobs.

Distance (Km)	0–10	10–20	20–30	30–40
Number of People	10	20	40	30

Solution:

The given distribution is grouped data and the variable involved is distance covered, while the number of people represents frequencies.

Distance (Km)	**Number of People** f	**Mid Points** x	fx
0–10	10	5	50
10–20	20	15	300
20–30	40	25	1000
30–40	30	35	1050
Total	∑f=100		∑fx=2400

$$\bar{x} = \frac{\sum fx}{N}$$

$$N = \sum f_i$$

=2400/100=24 Km.

3. If the mean of the following distribution is 9, find the value of p.

x	4	6	p + 7	10	15
f	5	10	10	7	8

Solution:

x_i	f_i	$x_i f_i$
4	5	20
6	10	60
p + 7	10	10(p + 7)
10	7	70
15	8	120

$\sum f_i = 5 + 10 + 10 + 7 + 8 = 40$, Given that $\bar{x} = 9$

$\sum f_i x_i = 270 + 10(p + 7)$

Mean = $\bar{x} = \frac{\sum fx}{N}$

$$9 = \frac{270 + 10(p + 7)}{40}$$

$\Rightarrow 270 + 10p + 70 = 9 \times 40$

$\Rightarrow 340 + 10p = 360$

$\Rightarrow 10p = 360 - 340$

$\Rightarrow 10p = 20$

$\Rightarrow p = 20/10$

$\Rightarrow p = 2$

4. The following table shows the number of plants in 20 houses in a group

Number of Plants	0 - 2	2 - 4	4 - 6	6 - 8	8 - 10	10 - 12	12 - 14
Number of Houses	1	2	2	4	6	2	3

Find the mean number of plants per house

Solution:

Number of Plant	**Number of Houses (f_i)**	**Class Mark (x_i)**	**$f_i x_i$**
0 - 2	1	1	1 × 1 = 1
2 - 4	2	3	2 × 3 = 6
4 - 6	2	5	2 × 5 = 10
6 - 8	4	7	4 × 7 = 28
8 - 10	6	9	6 × 9 = 54
10 - 12	2	11	2 × 11 = 22
12 -14	3	13	3 × 13 = 39

$\sum f_i = 1 + 2 + 2 + 4 + 6 + 2 + 3 = 20$

$\sum f_i x_i = 1 + 6 + 10 + 28 + 54 + 22 + 39 = 160$

Therefore, mean = $\bar{x} = \frac{\sum fx}{N}$

= 160/20 = 8 plants

Example:

The following data shows distance covered by 100 people to perform their routine jobs. Calculate the arithmetic mean by step-deviation method.

Distance (Km)	0–10	10–20	20–30	30–40
Number of People	10	20	40	30

Solution:

The given distribution is grouped data and the variable involved is distance covered, while the number of people represents frequencies.

Distance Covered in (Km)	**Number of People** f	**Mid Points** x	u=(x–15)/10	fu
0–10	10	5	–1	–10
10–20	20	15	0	0
20–30	40	25	+1	40
30–40	30	35	+2	60
Total	∑f=100			∑fu=90

A=15, ∑fu=90, ∑f=100 and h=10

$$\bar{x} = A + \frac{\sum fu}{N}.h$$

$$\bar{x} = 15 + \frac{90}{100}.10 = 24km$$

Example:

The following frequency distribution showing the marks obtained by 50 students in statistics at a certain college. Find the arithmetic mean using

(1) direct method

(2) short-cut method

(3) step-deviation.

Marks	20–29	30–39	40–49	50–59	60–69	70–79	80–89
Frequency	1	5	12	15	9	6	2

Solution:

			Direct Method	Short-Cut Method		Step-Deviation Method	
Marks	f	x	fx	D=x–A	fD	u=(x–A)/h	fu
20–29	1	24.5	24.5	–30	–30	–3	–3
30–39	5	34.5	172.5	–20	–100	–2	–10
40–49	12	44.5	534.5	–10	–120	–1	–12
50–59	15	54.5	817.5	0	0	0	0
60–69	9	64.5	580.5	10	90	1	9
70–79	6	74.5	447.5	20	120	2	12
80–89	2	84.5	169.5	30	60	3	6
Total	50		2745		20		2

1) Direct Method:

$$\bar{x} = \frac{\sum fx}{N}$$
$$N = \sum f_i$$

=2745/50=54.9 or 55Marks

(2) Short-Cut Method:

$$\bar{x} = A + \frac{\sum fD}{N}$$
$$D = x - A$$

Where A=54.5

=54.5+20/50=54.9Marks

(3) Step-Deviation Method:

$$\bar{x} = A + \frac{\sum fu}{N} . h$$

Where A=54.5 h=10

=54.5+(2/50)×10

=54.5+0.4=54.9 Marks

Merits and demerits of Arithmetic mean

Merits

1. It is rigidly defined.
2. It is easy to understand and easy to calculate.
3. If the number of items is sufficiently large, it is more accurate and more reliable.
4. It is a calculated value and is not based on its position in the series.
5. It is possible to calculate even if some of the details of the data are lacking.
6. Of all averages, it is affected least by fluctuations of sampling.
7. It provides a good basis for comparison.

Demerits

1. It cannot be obtained by inspection nor located through a frequency graph.
2. It cannot be in the study of qualitative phenomena not capable of numerical measurement i.e. Intelligence, beauty, honesty etc.,
3. It can ignore any single item only at the risk of losing its accuracy.
4. It is affected very much by extreme values.
5. It cannot be calculated for open-end classes.
6. It may lead to fallacious conclusions, if the details of the data from which it is computed are not given.

2. Median:

The median is a measure of central tendency, which denotes the value of the middle-most observation in the data.

For ungrouped data:

Median = $\left(\frac{n+1}{2}\right)^{th}$ observation, if n is odd.

Median = mean of $\left(\frac{n}{2}\right)^{th}$ observation and $\left(\frac{n}{2}+1\right)^{th}$ observation, if n is even.

Procedure:

Step 1: Arrange the given values in the ascending order.

Step 2: Find the number of observations in the given set of data. It is denoted by n.

Step 3: If n is odd, the median equals the $\left(\frac{n+1}{2}\right)^{th}$ observation.

Step 4: If n is even, then the median is given by the mean of $\left(\frac{n}{2}\right)^{th}$ observation and $\left(\frac{n}{2}+1\right)^{th}$ observation.

Problems :

1. Alex timed 21 people in the sprint race, to the nearest second:
   59, 65, 61, 62, 53, 55, 60, 70, 64, 56, 58, 58, 62, 62, 68, 65, 56, 59, 68, 61, 67

Solution:

Arranging the given values in the ascending order

53, 55, 56, 56, 58, 58, 59, 59, 60, 61, **61**, 62, 62, 62, 64, 65, 65, 67, 68, 68, 70

Median = $\left(\frac{n+1}{2}\right)^{th}$ observation, if n is odd.

$= \left(\frac{21+1}{2}\right)^{th} = 11^{th}$ observation

=61

Median=61

2. Find the median of the values **5, 7, 10, 20, 16, 12**

Solution:

Arrange the data in ascending order

5, 7, 10, 12, 16, 20

Since n=6 is even,

Median = mean of $\left(\frac{n}{2}\right)^{th}$ observation and $\left(\frac{n}{2}+1\right)^{th}$ observation, if n is even

= mean of 3rd and 4th observation

Median = $\frac{10+12}{2}=11$

Median = 11

3. Find the median of the values **4, 1, 8, 13, 11** **1,4,8,11,13**

4. Find the median of the following set of data

13, 8, 19, 30, 15, 21, 9, 5 **5,8,9,13,15,19,21,30**

For grouped data:

$$\text{Median} = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$

where

l= Lower class boundary of the median class

f= Frequency of the median class

N=∑f= Total frequency

c= Cumulative frequency of the class preceding the median class

h= Class interval size of the model class

Step 1: Make a table with 3 columns. First column for the class interval, second column for frequency f and the third column for cumulative frequency c.

Step 2: Write the class intervals and the corresponding frequency in the respective columns.

Step 3: Write the cumulative frequency in the column c. It is done by adding the frequency in each step

Step 4: Find the sum of frequencies N i.e., ∑f. It will be the same as the last number in the cumulative frequency column.

Step 5: Find N/2. Then find the class whose cumulative frequency is greater than and nearest to N/2. This is the median class.

Step 6: Now use the formula Median = $l + \frac{h}{f}\left(\frac{N}{2} - c\right)$ where

l= Lower class boundary of the model class

f= Frequency of the median class

N=∑f= Total frequency

c= Cumulative frequency of the class preceding the median class

h= Class interval size of the model class

Problems:

1.Calculate the median from the following data:

Group	60 – 64	65 – 69	70 – 74	75 – 79	80 – 84	85 – 89
Frequency	1	5	9	12	7	2

Solution:

Group	f	Cumulative Frequency	
60 – 64	1	1	
65 – 69	5	1+5=6	
70 – 74	9	6+9=15	
75 – 79	12	15+12=27	← Median class
80 – 84	7	27+7=34	
85 – 89	2	34+2=36	
	N=36		

$\because \left(\frac{N}{2}\right)^{th} = \frac{36}{2} = 18^{th}$

l= Lower class boundary of the model class= 74.5

f= Frequency of the median class = 12

N= 36

c= Cumulative frequency of the class preceding the median class= 15

h= Class interval size of the model class= 5

Median = $l + \frac{h}{f}\left(\frac{N}{2} - c\right)$

$= 74.5 + \frac{5}{12}\left(\frac{36}{2} - 15\right)$

$= 75.75$

Median =75.75

Median from Discrete Data

When the data follows a discrete set of values grouped by size, we use the formula $\left(\frac{N}{2}\right)^{th}$ item for finding the median. First form a cumulative frequency distribution, and the median is the value of x which corresponds to the cumulative frequency in which $\left(\frac{N}{2}\right)^{th}$ item lies.

Problem:

The following frequency distribution is classified according to the number of leaves on different branches. Calculate the median number of leaves per branch.

No of Leaves	1	2	3	4	5	6	7
No of Branches	2	11	15	20	25	18	10

Solution:

No of Leaves *X*	No of Branches *f*	Cumulative Frequency *C.*
1	2	2
2	11	13
3	15	28
4	20	48
5	25	73
6	18	91
7	10	101
Total	N=101	

Median = Size of $\left(\frac{N}{2}\right)^{th}$ item

$$= \left(\frac{101}{2}\right)^{th} = 50.5$$

Median= Value of x corresponding to just greater than cumulative frequency 50.5= 5

3. If the median of the following frequency distribution is 28.5, find the missing frequencies.

Class	0-10	10-20	20-30	30-40	40-50	50-60	Total
Frequency	f1	8	20	15	7	f2	60

Solution:

class	Frequency *f*	Cumulative Frequency *C.*
0-10	**f1**	**f1**
10-20	**8**	**f1+8**
20-30	**20**	**f1+28**
30-40	**15**	**f1+43**
40-50	**7**	**f1+50**
50-60	**f2**	**f1+f2+50**
Total	**N=60**	

N=f1+8+20+15+7+f2=60

f1+f2=60-50

⇒f1+f2=10

It is given that the median is 28.5.

Clearly, 28.5 lies in the median class 20-30. So, 20-30 is the median class.

l=20, f=20, c=f1+8, N=60, h=10

Median = $l + \frac{h}{f}\left(\frac{N}{2} - c\right)$

$$28.5 = 20 + \frac{10}{20}\left(\frac{60}{2} - (f1 + 8)\right)$$

$$28.5 - 20 = \frac{1}{2}(30 - f1 - 8)$$

$$8.5 * 2 = (22 - f1)$$

$$f1 = 22 - 17 = 5$$

$$f1 + f2 = 10$$

$$f2 = 10 - 5 = 5$$

$$\therefore f1 = 5,\ f2 = 5$$

Practice Problems:

1.The number of magazines read by a group of women in a week is recorded. If the median of the distribution is 2, find the value of x.

Number of magazines	0	1	2	3
Number of women	5	2	1	x

2. The shoe size of 155 people was recorded and the raw data was presented in the form of the following frequency table:

Size of shoe	4.0	4.5	5.0	5.5	6.0	6.5	7.0	7.5	8.0
Frequency	10	18	22	25	40	15	10	8	7

3. The marks obtained in English test by 17 students were recorded. What is the median marks of the students?

Marks out of 50	**Frequency**
0-10	2
10-20	4
20-30	5
30-40	4
40-50	2

Merits of Median

1. Median is not influenced by extreme values because it is a positional average.
2. Median can be calculated in case of distribution with open-end intervals.
3. Median can be located even if the data are incomplete.

Demerits of Median

1. A slight change in the series may bring drastic change in median value.
2. In case of even number of items or continuous series, median is an estimated value other than any value in the series.
3. It is not suitable for further mathematical treatment except its use in calculating mean deviation.
4. It does not take into account all the observations.

3. Mode:

The mode is the value that appears most often in a set of data i.e. the value whose frequency is maximum. The mode of a discrete probability distribution is the value x at which its probability mass function takes the maximum value.

Mode from Ungrouped Data

Mode is calculated from ungrouped data by inspecting the given data. We pick out the value which occurs the greatest number of times in the data.

Mode from Grouped Data

With frequency distribution with equal class interval sizes, the class which has the maximum frequency is called the model class.

$$\text{Mode} = l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} * h$$

l = Lower class boundary of the model class

f_m = Frequency of the model class (maximum frequency)

f_1 = Frequency preceding the model class frequency

f_2 = Frequency following the model class frequency

h = Class interval size of the model class

Mode from Discrete Data

When the data follows a discrete set of values, the mode may be found by inspection. The mode is the value of *X* corresponding to the maximum frequency.

Merits and Demerits of Mode

Merits	*Demerits*
1. Mode is readily comprehensible and easy to calculate. Like median, mode can be located in some cases merely by inspection.	1. Mode is ill-defined. It is not always possible to find a clearly defined mode. In some cases, we may come across distributions with two modes. Such distributions are called *bi-modal*. If a distribution has more than two modes, it is said to be *multimodal*.
2. Mode is not at all affected by extreme values.	2. It is not based upon all the observations.
3. Mode can be conveniently located even if the frequency distribution has class-intervals of unequal magnitude provided the modal class and the classes preceding and succeeding it are of the same magnitude. Open-end classes also do not pose any problem in the location of mode.	3. It is not capable of further mathematical treatment.
	4. As compared with mean, mode is affected to a greater extent, by fluctuations of sampling.

1. Example:

Find the mode of the values **5, 7, 2, 9, 7, 10, 8, 5, 7**

Solution:

The mode is **7** because it occurs the greatest number of times in the data.

2. Example:

The weights of 50 college students are given in the following table. Find the mode of the distribution.

Weight (Kg)	60 – 64	65 – 69	70 – 74	75 – 79	80 – 84
No of Students	5	9	16	12	8

Solution:

Weight (Kg)	No of Students f	Class Boundary
60 – 64	5	59.5 – 64.5
65 – 69	9	64.5 – 69.5
70 – 74	16	69.5 – 74.5
75 – 79	12	74.5 – 79.5
80 – 84	8	79.5 – 84.5

← Highest frequency 16

Mode= $l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} * h$

l = Lower class boundary of the model class = 69.5

f_m = Frequency of the model class (maximum frequency)= 16

f_1= Frequency preceding the model class frequency=9

f_2 = Frequency following the model class frequency= 12

h = Class interval size of the model class= 5

$$= 69.5 + \frac{16 - 9}{(16 - 9) + (16 - 12)} * 5$$

$$= 72.68$$

Mode= 72.68

Practice Problems:

1. Find the mode of the given data.

Class	Frequency
50-55	2
55-60	7
60-65	8
65-70	4

2. Alex timed 21 people in the sprint race, to the nearest second:

59, 65, 61, 62, 53, 55, 60, 70, 64, 56, 58, 58, 62, 62, 68, 65, 56, 59, 68, 61, 67

Find mode of the above distribution.

3. Find the mode of the data:

27,23,39,18,27,21,27,27,40,36,27

3. You grew fifty baby carrots using special soil. You dig them up and measure their lengths (to the nearest mm) and group the results:

Length (mm)	Frequency
150 - 154	5
155 - 159	2
160 - 164	6
165 - 169	8
170 - 174	9
175 - 179	11
180 - 184	6
185 - 189	3

Find mode of the above distribution.

4. Geometric mean:

The Geometric Mean (G.M.) of a set of n observations is the nth root of their product.

If $x_1, x_2, x_3, ---, x_n$ are n observations then

$$G.M = \sqrt[n]{x_1 \times x_2 \times x_3 \times \ldots \times x_n} = (x_1 \times x_2 \times x_3 \times \ldots \times x_n)^{\frac{1}{n}}$$

Taking the nth root of a number is difficult. Thus, the computation is done as

$$\log(G.M) = \log(x_1 \times x_2 \times x_3 \times \ldots \times x_n)^{\frac{1}{n}}$$

$$\log(G.M) = \frac{1}{n}\log(x_1 \times x_2 \times x_3 \times \ldots \times x_n)$$

$$= \frac{1}{n}(\log x_1 + \log x_2 + \ldots\ldots + \log x_n)$$

$$\log(G.M) = \frac{\sum_{i=1}^{n} \log x_i}{n}$$

$$G.M = anti\log\left(\frac{\sum_{i=1}^{n} \log x_i}{n}\right)$$

Example:

1. Find the geometric mean of the values 10, 5, 15, 8, 12.

Solution:

$$x_1 = 10, x_2 = 5, x_3 = 15, x_4 = 8 \; and \; x_5 = 12$$

Here n=5

$$G.M = \sqrt[n]{x_1.x_2.x_3....x_n}$$

$$= \sqrt[5]{10 \times 5 \times 15 \times 8 \times 12}$$

$$= (72000)^{\frac{1}{5}} = 9.36$$

$$G.M = 9.36$$

(OR)

x	log x
10	1.0000
5	0.6990
15	1.1761
8	0.9031
12	1.0792
Total	∑log x=4.8573

$$G.M = anti\log\left(\frac{\sum_{i=1}^{n}\log x_i}{n}\right)$$

$$= anti\log\left(\frac{4.8573}{5}\right)$$

$$= anti\log(0.9715) = 9.36$$

$$G.M = 9.36$$

2. Calculate the geometric mean of the annual percentage growth rate of profits in business corporate from the year 2000 to 2005 is given below

50, 72, 54, 82, 93

Solution:

x_i	50	72	54	82	93	Total
$\log x_i$	1.6990	1.8573	1.7324	1.9138	1.9685	9.1710

$$\text{G.M.} = \text{Antilog}\ \frac{\sum_{i=1}^{n} \log x_i}{n}$$

$$= \text{Antilog}\ \frac{9.1710}{5}$$

$$= \text{Antilog}\ 1.8342$$

$$\text{G. M.} = 68.26$$

Geometrical mean of annual percentage growth rate of profits is 68.26

Geometric mean for grouped data:

$$G.M = \sqrt[n]{x_1^{f_1} \times x_2^{f_2} \times x_3^{f_3} \times .... \times x_n^{f_n}}$$

or

$$G.M = anti\log\left(\frac{\sum_{i=1}^{n} f_i \log x_i}{N}\right)$$

1. Find the G.M for the following data, which gives the defective screws obtained in a factory.

Diameter (cm)	5	15	25	35
Number of defective screws	5	8	3	4

Solution:

x_i	f_i	$\log x_i$	$f_i \log x_i$
5	5	0.6990	3.4950
15	8	1.1761	9.4088
25	3	1.3979	4.1937
35	4	1.5441	6.1764
	N=20		23.2739

$$G.M = \text{Antilog}$$

$$= \text{Antilog} \frac{\sum_{i=1}^{n} f_i \log x_i}{N}$$

$$= \text{Antilog} \frac{23.2739}{20}$$

$$= \text{Antilog } 1.1637$$

$$G.M = 14.58$$

(c) G.M. for Continuous grouped data

1. The following is the distribution of marks obtained by 109 students in a subject in an institution. Find the Geometric mean.

Marks	4-8	8-12	12-16	16-20	20-24	24-28	28-32	32-36	36-40
No. of Students	6	10	18	30	15	12	10	6	2

Solution:

Marks	Mid point (x_i)	f_i	$\log x_i$	$f_i \log x_i$
4-8	6	6	0.7782	4.6692
8-12	10	10	1.0000	10.0000
12-16	14	18	1.1461	20.6298
16-20	15	30	1.2553	37.6590
20-24	22	15	1.3424	20.1360
24-28	26	12	1.4150	16.800
28-32	30	10	1.4771	14.7710
32-36	34	6	1.5315	9.1890
36-40	38	2	1.5798	3.1596
Total		**N =109**		**137.1936**

18

$$\text{G.M.} = \text{Antilog}\left[\frac{\sum_{i=1}^{n} f_i \log x_i}{N}\right]$$

$$= \text{Antilog}\left[\frac{137.1936}{109}\right] = \text{Antilog}\,[1.2587]$$

$$\text{G. M.} = 18.14$$

Merits and Demerits of Geometric Mean:

Merits	*Demerits*
1. It is rigidly defined. 2. It is based upon all the observations. 3. It is suitable for further mathematical treatment. 4. It is not affected much by fluctuations of sampling. 5. It gives comparatively more weight to small items.	1. Because of its abstract mathematical character, geometric mean is not easy to understand and to calculate for a non-mathematics person. 2. If any one of the observations is zero, geometric *mean* becomes zero and if any one of the observations is negative, geometric mean becomes imaginary regardless of the magnitude of the other items.

Practice problems:

1. For the frequency distribution of weights of sorghum ear-heads given in table below. Calculate the Geometric mean

Weights of ear heads (in g)	No of ear heads (f)
60-80	22
80-100	38
100-120	45
120-140	35
140-160	20
Total	**160**

2. Find the geometric mean of the following data

X	13	14	15	16	17
f	2	5	13	7	3

3) Find the geometrical mean of the number $3, 3^2, \ldots, 3^n$.

Harmonic mean:

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic average of the reciprocals of the given values.

$$If\ x_1, x_2, x_3, \ldots\ldots, x_n\ are\ n\ observations, then$$

$$H.M = \frac{n}{\sum_{i=1}^{n}\left(\frac{1}{x_i}\right)}$$

For frequency distribution,

$$H.M = \frac{N}{\sum_{i=1}^{n} f_i\left(\frac{1}{x_i}\right)}$$

Problems:

1. From the given data 5,10,17,24,30 calculate harmonic mean

Solution:

x	1/x
5	0.2
10	0.1
17	0.0588
24	0.0417
30	0.03333
Total	0.4338

$$H.M = \frac{n}{\sum_{i=1}^{n}\left(\frac{1}{x_i}\right)} = \frac{5}{0.4338} = 11.526$$

2. Number of tomatoes per plant are given below. Calculate the harmonic mean.

Number of tomatoes per plant	20	21	22	23	24	25
Number of plants	4	2	7	1	3	1

Solution

Number of tomatoes per plant (x)	No of plants(f)	$\frac{1}{x}$	$f\left(\frac{1}{x}\right)$
20	4	0.0500	0.2000
21	2	0.0476	0.0952
22	7	0.0454	0.3178
23	1	0.0435	0.0435
24	3	0.0417	0.1251
25	1	0.0400	0.0400
	18		0.8216

For frequency distribution,

$$H.M = \frac{N}{\sum_{i=1}^{n} f_i\left(\frac{1}{x_i}\right)}$$

$$= \frac{18}{0.8216} = 21.91$$

H.M=21.91

3. Calculate the harmonic mean for the data given below:

Marks	30–39	40–49	50–59	60–69	70–79	80–89	90–99
f	2	3	11	20	32	25	7

Solution:

Marks	x	f	f/x
30–39	34.5	2	0.0580
40–49	44.5	3	0.0674
50–59	54.5	11	0.2018
60–69	64.5	20	0.3101
70–79	74.5	32	0.4295
80–89	84.5	25	0.2959
90–99	94.5	7	0.0741
Total		∑f=100	∑(f/x)=1.4368

For frequency distribution,

$$H.M = \frac{N}{\sum_{i=1}^{n} f_i \left(\frac{1}{x_i} \right)}$$

$$= \frac{100}{1.4368}$$
$$= 69.60$$
$$\therefore H.M = 69.6$$

Practice problems:

1. The following data is obtained from the survey. Compute H.M

Speed of the car	130	135	140	145	150
No of cars	3	4	8	9	2

2. Find the harmonic mean of the following distribution of data

Dividend yield (percent)	2 – 6	6 – 10	10 – 14
No. of companies	10	12	18

3. For the following frequency distribution find the

(i) Mean

(ii) Median

(iii) Mode

(iv) Harmonic mean

(iv) Geometric mean

Weight of earheads in gms	No. of earhead
40 - 60	6
60 - 80	8
80 – 100	35
100 -120	55
120 -140	30
140 – 160	15
160 – 180	12
180 – 200	9

Merits of H.M

1. It is rigidly defined.
2. It is defined on all observations.
3. It is amenable to further algebraic treatment.
4. It is the most suitable average when it is desired to give greater weight to smaller observations and less weight to the larger ones.

Demerits of H.M

1. It is not easily understood.
2. It is difficult to compute.
3. It is only a summary figure and may not be the actual item in the series
4. It gives greater importance to small items and is therefore, useful only when small items have to be given greater weightage.
5. It is rarely used in grouped data.

Measures of Variability:

A measure of variability is a summary statistic that represents the amount of dispersion in a dataset. While a measure of central tendency describes the typical value, measures of variability define how far away the data points tend to fall from the center.

There are many ways to describe variability or spread including:

1. Range
2. Interquartile range (IQR)
3. Variance and Standard Deviation

Range:

The range is the difference in the maximum and minimum values of a data set. The maximum is the largest value in the dataset and the minimum is the smallest value. The range is easy to calculate but it is very much affected by extreme values.

Range=maximum−minimum

Interquartile Range (IQR):

The interquartile range is the difference between upper and lower quartiles and denoted as **IQR**.

IQR=Q3−Q1=upper quartile−lower quartile=75th percentile−25th percentile

Range and Interquartile range example:

1. You have 8 data points from Sample A.

Data (minutes)	72	110	134	190	238	287	305	324

The highest value (*H*) is **324** and the lowest (*L*) is **72**.

$R = H - L$

$R = 324 - 72 = \mathbf{252}$

To find the interquartile range of 8 data points, first find the values at Q1 and Q3.

Multiply the number of values in the data set (8) by 0.25 for the 25th percentile (Q1) and by 0.75 for the 75th percentile (Q3).

Q1 position: 0.25 x 8 = 2

Q3 position: 0.75 x 8 = 6

Q1 is the value in the 2nd position, which is **110**. Q3 is the value in the 6th position, which is **287**.

IQR = Q3 – Q1

IQR = 287 – 110 = **177**

The interquartile range of given data is **177 minutes**.

Variance formula for samples

Formula	Explanation
$s^2 = \frac{\Sigma (X - \bar{x})^2}{n - 1}$	• s^2 = sample variance • Σ = sum of… • X = each value • $\bar{x}$ = sample mean • n = number of values in the sample

Standard deviation:

Square root of variance is called standard deviation.

Standard deviation formula for Population:

Formula	Explanation
$\sigma = \sqrt{\frac{\Sigma (X-\mu)^2}{N}}$	•σ = population standard deviation •∑ = sum of… •*X* = each value •μ = population mean •*N* = number of values in the population

Standard deviation formula for samples:

Formula	Explanation
$s = \sqrt{\frac{\Sigma (X-\bar{x})^2}{n-1}}$	•*s* = sample standard deviation •∑ = sum of… •*X* = each value •$\bar{x}$ = sample mean •*n* = number of values in the sample

Problems:

1. Calculate the variance and standard deviation for these population of final exam scores.

61, 67, 73, 75, 76, 78, 81

Solution:

$$\mu = \frac{sum\ of\ values}{no.of\ values}$$

$$= \frac{61+67+73+75+76+78+81}{7}$$

$$= 73$$

$$Variance(\sigma^2) = \frac{\sum_{i=1}^{n}(x_i - \mu)^2}{N}$$

$$= \frac{(61-73)^2+(67-73)^2+(73-73)^2+(75-73)^2+(76-73)^2+(78-73)^2+(81-73)^2}{7}$$

$$= \frac{(-12)^2+(-6)^2+(0)^2+(2)^2+(3)^2+(5)^2+(8)^2}{7}$$

$$= \frac{144+36+4+9+25+64}{7}$$

$$= 40.28$$

$$Variance = 40.28$$

Standard deviation $= \sqrt{Variance}$

$$= \sqrt{40.28} = 6.3467$$

Standard deviation=6.3467

2.Find the variance, and the standard deviation for the following sample.

2 −3 6 0 3 1

Solution:

$$\bar{x} = \frac{2-3+6+0+3+1}{6}$$

$$= \frac{9}{6} = 1.5$$

$$\bar{x} = 1.5$$

$$S^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

$$= \frac{(2-1.5)^2 + (-3-1.5)^2 + (6-1.5)^2 + (0-1.5)^2 + (3-1.5)^2 + (1-1.5)^2}{6-1}$$

$$= \frac{(0.5)^2 + (-4.5)^2 + (4.5)^2 + (-1.5)^2 + (1.5)^2 + (-0.5)^2}{5}$$

$$= \frac{.25 + 20.25 + 20.25 + 2.25 + 2.25 + 0.25}{5}$$

$$= 9.1$$

$$Variance = 9.1$$

$$S\tan dard\ deviation = \sqrt{Variance}$$

$$= \sqrt{9.1} = 3.017$$

$$S\tan dard\ deviation = 3.017$$

3. Find mean , variance and standard deviation of the given dataset I

Data Set I:	46	37	40	33	42	36	40	47	34	45

Variance and standard deviation for grouped data:

$$\sigma^2 = \frac{\sum_{i=1}^{n} f_i (x_i - A)^2}{N} (or) \frac{\sum_{i=1}^{n} f_i d_i^{\,2}}{N} \; where \; d = x - A$$

A is arbitrary value.

$$where \; N = \sum f_i$$

$$S\tan dard \; deviation = \sqrt{Variance}$$

1. Calculate the variance of the following table.

x	2	4	6	8	10	12	14	16
f	4	4	5	15	8	5	4	5

Solution : let A=8

x	f	d = x-8	d^2	fd^2
2	4	2-8 = -6	36	144
4	4	4-8 = -4	16	64
6	5	6-8 = -2	4	20
8	15	8-8 = 0	0	0
10	8	10-8 = 2	4	32
12	5	12-8 = 4	16	80
14	4	14-8 = 6	36	144
16	5	16-8 = 8	64	320

$\Sigma f = 50$ and $\Sigma fd^2 = 804$

$$\sigma^2 = \frac{\sum_{i=1}^{n} f_i d_i^{\,2}}{N}$$

= (804/50)

= 16.08

Standard deviation= $\sqrt{Variance}$

$$= \sqrt{16.08} = 4.009$$

Standard deviation=4.009

Question 2 :

Find the variance and standard deviation of the following distribution

Class interval	Frequency
20-24	15
25-29	25
30-34	28
35-39	12
40-44	12
45-49	8

Solution :Let A=32

Here we consider the first data that is class interval as (x) and no of frequency as (f).

Class interval	x	f	d = x-32	d^2	fd^2
20-24	22	15	22-32 = -10	100	1500
25-29	27	25	27-32 = -5	25	625
30-34	32	28	32-32 = 0	0	0
35-39	37	12	37-32 = 5	25	300
40-44	42	12	42-32 = 10	100	1200
45-49	47	8	47-32 = 15	225	1800

$\Sigma fd^2 = 5425$ and $\Sigma f = 100$

$$\sigma^2 = \frac{\sum_{i=1}^{n} f_i d_i^{\,2}}{N}$$

= (5425/100)

Variance= 54.25

Standard deviation= $\sqrt{\text{var}\,iance} = \sqrt{54.25}$

=7.37

Standard deviation =7.37

ADITYA UNIVERSITY

Sampling Theory

Contents

Introduction – Population and samples – Sampling distribution of Means and Variance – Central limit theorem (without proof)-Point and Interval estimations – Maximum error of estimate.

Sampling:

Sampling is a method that allows us to get information about the population based on the statistics from a subset of the population (sample), without having to investigate every individual.

Sampling Frame – It is a list of items or people forming a population from which the sample is taken.

Population(or Universe): A collection or aggregate or totality of persons, things or objects or statistical data under study is known as population.

Size of the Population: Number of units in the population is known as size of the population. It is denoted by N.

Finite Population: The number of units in the population is finite then it is known as finite population.

Infinite Population: : The number of units in the population is infinite then it is known as infinite population.

Sample: A portion(or subset) of the population is known as sample.

Size of the sample: Number of units in the sample is known as size of the sample. It is denoted by n.

Large sample: If the size of the sample is greater than 30 then it is known as large sample.

Small sample: If the size of the sample is less than 30 then it is known as small sample.

Parameter: The values that are obtained from the population data or a number that describes the property of the population is known as parameters

Statistic: The values that are obtained from the sample data or a number that describes the property of the sample is known as Statistics.

Probability Sampling: In probability sampling, every element of the population has an equal chance of being selected. Probability sampling gives us the best chance to create a sample that is truly representative of the population

Non-Probability Sampling: In non-probability sampling, all elements do not have an equal chance of being selected. Consequently, there is a significant risk of ending up with a non-representative sample which does not produce generalizable results.

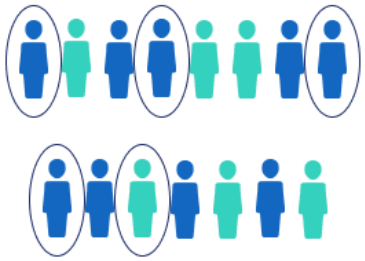
Probabilistic sampling methods:

1. Simple random sampling
2. Systematic sampling
3. Stratified sampling
4. Cluster sampling

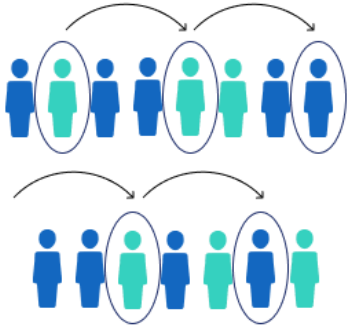
Non-Probabilsitic sampling methods:

1. Purposive or Judgement sampling
2. Sequential sampling

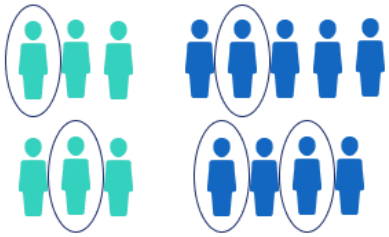
Simple random sample



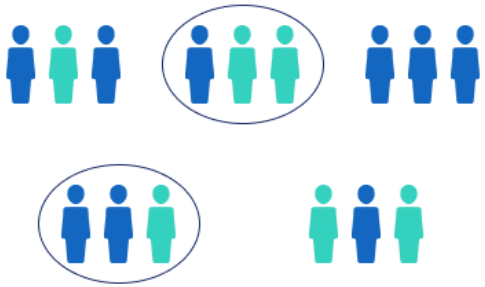
Systematic sample



Stratified sample



Cluster sample



Scribbr

Probabilistic sampling methods:

1. Simple random sampling

In a simple random sample, every member of the population has an equal chance of being included in the sample.. Your sampling frame should include the whole population. To conduct this type of sampling, tools like random number generators or other techniques are used that are based entirely on chance.

Ex: 1. selecting randomly 20 words from a dictionary is a random sample

2. Choosing 10 patients from a hospital in order to test the efficacy of a certain nely invented drug.

2. Systematic sampling or Quasi-random sampling:

In this method the sample is formed by some systematic manner by taking items at regular intervals.

All the units of the population are arranged in some order. Then from the first k items, one unit is selected at random. This unit and every kth unit of the serially listed population combined together constitute a systematic sample. This type of sampling is known as systematic sampling.

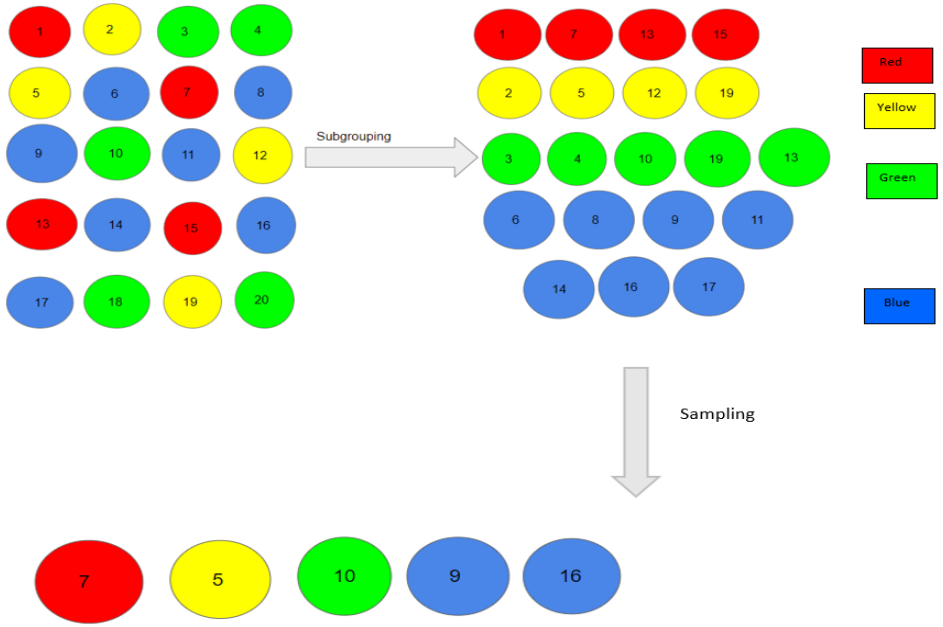
The difference between random sampling and systematic sampling is in case of random sampling all units are chosen randomly, whereas in case of systematic sampling only first unit is chosen at random.

3. Stratified sampling or stratified random sampling:

This method is useful when the population is heterogeneous.

In this type of sampling, the population is first sub-divided into several groups called strata according to some relevant characteristics(e.g. gender, age range, income bracket, job role). so that each stratum is more or less homogeneous. Each stratum is called sub population.

Then a small sample is select from each stratum at random. All the sub samples are combined to form the stratified sample which represents the population properly.



4. Cluster sampling

Cluster sampling also involves dividing the population into subgroups, but each subgroup should have similar characteristics to the whole sample. Instead of sampling individuals from each subgroup, you randomly select entire subgroups.

If it is practically possible, you might include every individual from each sampled cluster. If the clusters themselves are large, you can also sample individuals from within each cluster using one of the techniques above.

This method is good for dealing with large and dispersed populations, but there is more risk of error in the sample, as there could be substantial differences between clusters. It's difficult to guarantee that the sampled clusters are really representative of the whole population.

Non-probabilistic sampling methods:

1. **Purposive or Judgement sampling:**

When the choice the individual items of a sample entirely depends on the individual judgement of the investigator, it is called a purposive or Judgement sampling.

In this method, the members constituting the sample are chosen not according to some definite scientific procedure, but according to convenience and personal choice of the individual who selects the sample.

In this type, the investigator must have a good deal of experience and a thorough knowledge of the population.
Purposive selection is always subject to some kind of bias.
This method is suitable when the sample is small.

2. Sequential sampling:

It consists of a sequence of samples drawn one after another from the population depending on the results of the previous samples.
If the result of the first sample leads to a decision which is not acceptable, the lot from which the sample was drawn is rejected. Then a second ssample is drawnand as before if required, a third sample is drawn to arrive at a final decision to accept or reject the lot.
If the first sample is acceptable then there is no need to draw new sample.
It is widely used in Statistical quality control in factories engaged in mass production and other areas.

Population and Samples:

1. Sample mean:

The sample mean of a set of observations is the sum of observations divided by the number of observations in the data. It is denoted by $\bar{x}$

For n values in a set of data namely as $x_1, x_2, x_3, \ldots x_n$, the sample mean of data is given as:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n}$$

or $$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

Population mean :-

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

Variance:

Variance is the average squared difference of the values from the mean

Variance reflects the degree of spread in the data set. The more spread the data, the larger the variance is in relation to the mean.

Variance formula for populations

Formula	Explanation
$\sigma^2 = \frac{\Sigma (X - \mu)^2}{N}$	•σ^2 = population variance •Σ = sum of… •*X* = each value •μ = population mean •*N* = number of values in the population

Variance formula for samples

Formula	Explanation
$s^2 = \frac{\Sigma (X - \bar{x})^2}{n - 1}$	• s^2 = sample variance • Σ = sum of… • X = each value • $\bar{x}$ = sample mean • n = number of values in the sample

Standard deviation:

Square root of variance is called standard deviation.

Standard deviation formula for Population:

Formula	Explanation
$\sigma = \sqrt{\frac{\Sigma (X-\mu)^2}{N}}$	•σ = population standard deviation •∑ = sum of… •*X* = each value •μ = population mean •*N* = number of values in the population

Standard deviation formula for samples:

Formula	Explanation
$s = \sqrt{\frac{\Sigma (X-\bar{x})^2}{n-1}}$	•*s* = sample standard deviation •∑ = sum of… •*X* = each value •$\bar{x}$ = sample mean •*n* = number of values in the sample

Sampling Distributions: The probability distribution of a sample statistic is often called as sampling distribution of the statistic.

The standard deviation of the sampling distribution of a statistic is called **Standard Error(S.E)**

The mean of the sampling distribution of means, denoted by $\mu_{\bar{x}}$, is given by $E(\bar{X}) = \mu_{\bar{x}} = \mu$ where μ is the mean of the population. $\mu_{\bar{x}} = \mu$

If a population is infinite or if sampling is with replacement, then the variance of the sampling distribution of means, denoted by $\sigma_{\bar{x}}^2$ is given by $E[(\bar{X} - \mu)^2] = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ where σ^2 is the variance of the population.

$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ (with replacement)

$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$ (without replace)

If the population is of siqe N, if sampling is without replacement, and if the sample size is

$n \leq N$ then $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right)$

The factor $\left(\frac{N-n}{N-1}\right)$ is called the finite population correction factor, is close to 1 (and can be omitted for most practical purposes) unless the samples constitutes a substantial portion of the population.

Note: No. of samples of size n can be drawn from a population of size N

a) with replacement $= N^n$

b) without replacement $= {}^N C_n$.

Sampling distribution of means:-

$S_1 \to \bar{x}_1$

$S_2 \to \bar{x}_2$

$S_3 \to \bar{x}_3$

$S_4 \to \bar{x}_4$

$\vdots$

$S_n \to \bar{x}_n$

$\bar{X} = \{\bar{x}_1, \bar{x}_2, \bar{x}_3, \cdots, \bar{x}_n\}$

mean of sampling distri of means

$$\mu_{\bar{x}} = \frac{\bar{x}_1 + \bar{x}_2 + \cdots + \bar{x}_n}{n} = \frac{\sum_{i=1}^{n} \bar{x}_i}{n}$$

$$\sigma_{\bar{x}}^2 = \frac{(\bar{x}_1 - \mu_{\bar{x}})^2 + (\bar{x}_2 - \mu_{\bar{x}})^2 + \cdots + (\bar{x}_n - \mu_{\bar{x}})^2}{n}$$

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{n} (\bar{x}_i - \mu_{\bar{x}})^2}{n}$$

$\mu_{\bar{x}} = \mu$

mean of sampling distributions of means = mean of pop

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \quad \text{(with replacement)}$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right) \quad \text{(without replacement)}$$

PROBLEMS:1. A population consists of five numbers 2,3,6,8 and 11. Consider all possible samples of size two which can be drawn with replacement from this population. Find

i) The mean of the population
ii) The standard deviation of the population
iii) The mean of the sampling distribution of means
iv) The standard deviation of the sampling distribution of means

Solution: Given that N=5, n=2 and

i. Mean of the population

$$\mu = \frac{\sum x_i}{N} = \frac{2 + 3 + 6 + 8 + 11}{5} = \frac{30}{5} = 6$$

ii. Variance of the population

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N} = \frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}$$

$$= \frac{16 + 9 + 0 + 4 + 25}{5}$$

$$= 10.8$$

$$\therefore \sigma = 3.29$$

Sampling with replacement(infinite population):
The total number of samples with replacement is

$$N^n = 5^2 = 25$$

There 25 samples can be drawn

$$\begin{Bmatrix} (2,2) & (2,3) & (2,6) & (2,8) & (2.11) \\ (3,2) & (3,3) & (3,6) & (3,8) & (3,11) \\ (6,2) & (6,3) & (6.6) & (6,8) & (6,11) \\ (8,2) & (8,3) & (8,6) & (8,8) & (8,11) \\ (11,2) & `(11,3) & (11,6) & (11,8) & (11,11) \end{Bmatrix}$$

The sample means are

Sampling list of means =

$$\begin{Bmatrix} 2 & 2.5 & 4 & 5 & 6.5 \\ 2.5 & 3 & 4.5 & 5.5 & 7 \\ 4 & 4.5 & 6 & 7.0 & 8.5 \\ 5 & 5.5 & 7 & 8 & 9.5 \\ 6.5 & 7 & 8.5 & 9.5 & 11 \end{Bmatrix}$$

iii. The mean of the sampling distribution of means is

$$\mu_{\bar{x}} = \frac{2 + 2.5 + 4 + 5 + 6.5 + - - - - + 11}{25}$$

$$= 6$$

iv. The standard deviation of the sampling distribution of means

$$\sigma_{\bar{x}}^2 = \frac{(2-6)^2 + (2.5-6)^2 + - - - - + (11-6)^2}{25}$$

$$= 5.40$$

$$\sigma_{\bar{x}} = 2.32$$

(2) $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{3} = \frac{10.8}{2} = 5.4$

1) (b) <u>without</u> <u>replacement</u>

No of samples of size 2 that can be drawn from population of size 5 without replacement $= {}^{N}C_{n}$

$= {}^{5}C_{2} = \frac{5!}{3!\,2!} = \frac{5 \times 4 \times 3!}{3! \times 2} = 10$ samples

population is 2, 3, 6, 8, 11 2, 3, 6, 8, 11

Samples are (2,3) (2,6) (2,8) (2,11)
(3,6) (3,8) (3,11)
(6,8) (6,11) and (8,11)

Sampling distribution of means $(\bar{x}) = \{2.5, 4, 5, 6.5, 4.5, 5.5, 7, 7, 8.5, 9.5\}$

Mean of sampling distribution of means $[\mu_{\bar{x}}]$

$= (2.5+4+5+6.5+4.5+5.5+7+7+8.5+9.5)/10 = 6$

variance of sampling distribution of means

$$\sigma_{\bar{x}}^2 = \frac{\sum (\bar{x}_i - \mu_{\bar{x}})^2}{N}$$

$$= \frac{(2.5-6)^2 + (4-6)^2 + (5-6)^2 + (6.5-6)^2 + (4.5-6)^2 + (5.5-6)^2 + (7-6)^2 + (7-6)^2 + (8.5-6)^2 + (9.5-6)^2}{10}$$

$\sigma_{\bar{x}}^2 = 4.05$

standard deviation $\sigma_{\bar{x}} = \sqrt{4.05} = 2.01$

(ii)

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}\left(\frac{N-n}{N-1}\right) = \frac{10.8 \times (5-2)}{2 \times (5-1)} = \frac{10.8 \times 3}{2 \times 4}$$

$$= 4.05.$$

2. A population consists of five numbers4, 8, 12, 16, 20, 24. Consider all possible samples of size two which can be drawn without replacement from this population. Find

i) The mean of the population
ii) The standard deviation of the population
iii)The mean of the sampling distribution of means
iv) the standard deviation of the sampling distribution of means

Solution: Given that N=6, n=2 and

i. Mean of the population

$$\mu = \sum \frac{x_i}{N} = \frac{4 + 8 + 12 + 16 + 20 + 24}{6} = \frac{84}{6} = 14$$

ii. Variance of the population

$$\sigma^2 = \sum \frac{(x_i - \bar{x})^2}{N} = \frac{(4 - 14)^2 + (8 - 14)^2 + (12 - 14)^2 + (16 - 14)^2 + (20 - 14)^2 + (24 - 14)^2}{6}$$

$$= \frac{100 + 36 + 4 + 4 + 36 + 100}{6}$$

$$= 46.67 \qquad \sigma = 3.29$$

Sampling without replacement (finite population):

The total number of samples without replacement is $N_{c_n} = 6_{c_2} = 15$

There 15 samples can be drawn

$$\left\{\begin{array}{lllll} (4,8) & (4,12) & (4,16) & (4,20) & (4,24) \\ (8,12) & (8,16) & (8,20) & (8,24) & \\ (12,16) & (12,20) & (12,24) & & \\ (16,20) & (16,24) & & & \\ (20,24) & & & & \end{array}\right\}$$

The sample means are

$$\left\{\begin{array}{lllll} 6 & 8 & 10 & 12 & 14 \\ 10 & 12 & 14 & 16 & \\ 14 & 16 & 18 & & \\ 18 & 20 & & & \\ 22 & & & & \end{array}\right\}$$

iii. The mean of the sampling distribution of means is

$$\mu_{\bar{x}} = \frac{6 + 8 + 10 + 12 - - - - - 20 + 22}{15}$$

$$= \mathbf{14}$$

iv. The standard deviation of the sampling distribution of means

$$\sigma^2_{\bar{x}} = \frac{(6 - 14)^2 + (8 - 14)^2 + - - - - + (22 - 14)^2}{15}$$

$$= 18.67$$

$$\sigma_{\bar{x}} = 4.32$$

3. If the population is 3,6,9,15,27

a) List all possible samples of size 3 that can be taken without replacement from the population

b) Calculate the mean and variance of the population

c) Find the mean and standard deviation of the sampling distribution of means.

Solution:

N=5,n=3

No. of samples that can be drawn without replacement= $N_{c_n} = 5_{c_3} = 10$

a) The samples are

(3,6,9),(3,6,15),(3,6,27),(3,9,15),(3,9,27),(3,15,27)(6,9,15),(6,9,27),(6,15,27), (9,15,27)

b) Mean of the population $= \frac{\sum_{i=1}^{5} x_i}{N}$ (μ)

$$= \frac{3+6+9+15+27}{5} = 12$$

Variance of population $= \frac{\sum (x_i - \mu)^2}{N}$

$$= \frac{(3-12)^2 + (6-12)^2 + (9-12)^2 + (15-12)^2 + (27-12)^2}{5}$$

$$= \frac{360}{5} = 72$$

$S.D = \sqrt{72} = 8.485$

c) Sampling distribution of mean:

$\{6, 8, 12, 9, 13, 15, 10, 14, 16, 17\}$

Mean of sampling distribution of means $(\mu_{\bar{x}})$

$$= \frac{6+8+12+9+13+15+10+14+16+17}{10} = \frac{120}{10} = 12$$

variance of sampling distribution of means $(\sigma^2_{\bar{x}})$

$$= \frac{(6-12)^2 + (8-12)^2 + (12-12)^2 + (9-12)^2 + (13-12)^2 + (15-12)^2 + (10-12)^2 + (14-12)^2 + (16-12)^2 + (17-12)^2}{10}$$

$= 13.3$

$\therefore \sigma_{\bar{x}} = \sqrt{13.3} = 3.65$

3, 6, 9, 15, 27

(3, 6, 9) (3, 6, 15) (3, 6, 27) (3, 9, 15) (3, 9, 27)

(3, 15, 27) (6, 9, 15) (6, 9, 27) (6, 15, 27) (9, 15, 27)

15

sampling distribution of means

$= \{6, 8, 12, 9, 13, 15, 10, 14, 16, 17\}$

Mean of sampling dist of means ($\mu_{\bar{x}}$)

$$= \frac{6+8+12+9+13+15+10+14+16+17}{10} = 12$$

$$\sigma_{\bar{x}}^2 = \frac{\sum_{i=1}^{10}(\bar{x}_i - \mu_{\bar{x}})^2}{n} = \frac{(6-12)^2 + (8-12)^2 + (12-12)^2 + \cdots + (16-12)^2 + (17-12)^2}{10}$$

$$= 13.3$$

$$\sigma_{\bar{x}} = \sqrt{13.3} = 3.65$$

Exercise Problems:

4. Samples of size 2 are taken from the population 1, 2, 3, 4, 5, 6. Which can be drawn without replacement? Find

i) The mean of the population

ii) The standard deviation of the population

iii) The mean of the sampling distribution of means

iv) The standard deviation of the sampling distribution of means

Central limit theorem:

If $\bar{x}$ be the mean of a random sample of size n drawn from a population having mean μ and standard deviation σ, then the standardized sample mean is

$$z = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$$

is a random variable whose distribution function approaches that of the standard normal distribution N(z;0,1) as $n \to \infty$

Standard error formulas:

Standard error of sample mean $\bar{x}$ or S.E of $\bar{x}$ = $\frac{\sigma}{\sqrt{n}}$

Standard error of sample proportion p or S.E of p = $\sqrt{\frac{PQ}{n}}$

Standard error of sample standard deviation = $\frac{\sigma}{\sqrt{2n}}$ (s)

4. S.E of difference of sample means i.e., $(\bar{x}_1 - \bar{x}_2)$

$$= \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

5. S.E of difference of proportions i.e., $(p_1 - p_2)$

$$= \sqrt{\frac{P_1Q_1}{n_1} + \frac{P_2Q_2}{n_2}}$$

6. S.E of difference of standard deviations $(s_1 - s_2)$

$$= \sqrt{\frac{\sigma_1^2}{2n_1} + \frac{\sigma_2^2}{2n_2}}$$

Note:

1) For a finite population of size N, when a sample is drawn without replacement

(i) S.E of sample mean $= \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}}$.

(ii) S.E of sample proportion $= \sqrt{\frac{PQ}{n}} \cdot \sqrt{\frac{N-n}{N-1}}$.

where $\sqrt{\frac{N-n}{N-1}}$ is known as finite population correction factor.

Sampling distribution of differences and sums :-

1. $\mu_{S_1 - S_2} = \mu_{S_1} - \mu_{S_2}$

2. $\sigma_{S_1 - S_2} = \sqrt{\sigma_{S_1}^2 + \sigma_{S_2}^2}$

3. $\mu_{S_1 + S_2} = \mu_{S_1} + \mu_{S_2}$

4. $\sigma_{S_1 + S_2} = \sqrt{\sigma_{S_1}^2 + \sigma_{S_2}^2}$

5. $\mu_{\bar{x}_1 + \bar{x}_2} = \mu_{\bar{x}_1} + \mu_{\bar{x}_2} = \mu_1 + \mu_2$

6. $\sigma_{\bar{x}_1 + \bar{x}_2} = \sqrt{\sigma_{\bar{x}_1}^2 + \sigma_{\bar{x}_2}^2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

7. $\mu_{P_1 - P_2} = \mu_{P_1} - \mu_{P_2} = P_1 - P_2$

8. $\sigma_{P_1 - P_2} = \sqrt{\sigma_{P_1}^2 + \sigma_{P_2}^2} = \sqrt{\frac{P_1 Q_1}{n_1} + \frac{P_2 Q_2}{n_2}}$

1) Let $u_1 = (3,7,8)$, $u_2 = (2,4)$. Find

a) μ_{u_1} b) μ_{u_2} c) Mean of sampling distribution of the difference of means $\mu_{u_1 - u_2}$

d) σ_{u_1} e) σ_{u_2}

f) the standard deviation of the differences of means $\sigma_{u_1 - u_2}$.

Soln :- a) $\mu_{u_1} = \frac{3+7+8}{3}$

$= 18/3 = 6$

b) $\mu_{u_2} = \frac{2+4}{2} = \frac{6}{2} = 3$

$\therefore \mu_{u_2} = 3$

d) $u_1 = (3, 7, 8)$

$\mu_{u_1} = 6$

$$\sigma^2_{u_1} = \frac{(3-6)^2 + (7-6)^2 + (8-6)^2}{3}$$

$$= \frac{9+1+4}{3} = \frac{14}{3} = 4.667 \qquad \therefore \sigma^2_{u_1} = 4.667$$

$$\sigma_{u_1} = \sqrt{4.667} = 2.16$$

e) $u_2 = (2, 4)$, $\mu_{u_2} = 3$

$$\sigma^2_{u_2} = \frac{(2-3)^2 + (4-3)^2}{2}$$

$$= \frac{1+1}{2} = 1$$

$\therefore \sigma_{u_2} = 1.$

c) $u_1 = (3, 7, 8)$ $u_2 = (2, 4)$

$u_1 - u_2 = (1, -1, 5, 3, 6, 4)$

Mean of sampling distribution of differences $u_1 - u_2$

$$\mu_{u_1 - u_2} = \frac{1 + (-1) + 5 + 3 + 6 + 4}{6}$$

$$= \frac{18}{6} = 3$$

$$\therefore \mu_{u_1 - u_2} = 3$$

(or) $\mu_{u_1 - u_2} = \mu_{u_1} - \mu_{u_2}$

$= 6 - 3 = 3$.

$\therefore \mu_{u_1 - u_2} = 3$

f) Variance of sampling distribution of differences $u_1 - u_2$ i.e., $\mu_{u_1 - u_2} = 3$, $u_1 - u_2 = \{1, -1, 5, 3, 6, 4\}$

$$\sigma^2_{u_1-u_2} = \frac{(1-3)^2+(-1-3)^2+(5-3)^2+(3-3)^2+(6-3)^2+(4-3)^2}{6}$$

$$= \frac{4+16+4+0+9+1}{6}$$

$$= \frac{34}{6} = 5.667$$

Standard deviation $\sigma_{u_1-u_2} = \sqrt{\text{variance}}$

$$= \sqrt{5.667}$$

$$= 2.38$$

(or) $\sigma_{u_1-u_2} = \sqrt{\sigma_{u_1}^2+\sigma_{u_2}^2} = \sqrt{4.667+1} = \sqrt{5.667} = 2.38$

2) Find the value of the finite population correction factor for $n = 10$ & $N = 1000$.

Soln :- $N = 1000$

$n = 10$

$$\text{Correction factor} = \frac{N-n}{N-1} = \frac{1000-10}{1000-1}$$

$$= \frac{990}{999} = 0.991.$$

3) The variance of a population is 2. The size of the sample collected from the population is 169. What is standard error of mean?

Soln :- $\sigma^2 = 2$, $n = 169$

$$\text{S.E of } \bar{x} = \sqrt{\frac{\sigma^2}{n}} = \sqrt{\frac{2}{169}} = 0.185$$

4) when a sample is taken from an infinite population what happen to the standard error of mean if the sample size is decreased from 800 to 200.

Soln :- standard error for sample mean $\bar{x} = \frac{\sigma}{\sqrt{n}}$

If $n_1 = 800$ $S.E_1 = \frac{\sigma}{\sqrt{800}} = \frac{\sigma}{\sqrt{4 \times 200}} = \frac{\sigma}{2\sqrt{200}}$

If $n_2 = 200$

$$SE_2 = \frac{\sigma}{\sqrt{200}}$$

$$S.E_1 = \frac{1}{2}\left(\frac{\sigma}{\sqrt{200}}\right) = \frac{1}{2} \cdot SE_2$$

$$SE_2 = 2 \cdot SE_1$$

∴ If sample size decreased from 800 to 200, then S.E will be multiplied by 2.

5.What is the effect on standard error, if a sample is taken from an infinite population of sample size is increases from 400 to 900.

Solution:

$n_1 = 400, n_2 = 900$ S.E of sample mean $\bar{x} = \frac{\sigma}{\sqrt{n}}$

$$S.E_1 = \frac{\sigma}{\sqrt{n_1}} = \frac{\sigma}{\sqrt{400}} = \frac{\sigma}{20}$$

$$S.E_1 = \frac{\sigma}{20}$$

$$S.E_2 = \frac{\sigma}{\sqrt{n_2}} = \frac{\sigma}{\sqrt{900}} = \frac{\sigma}{30}$$

$$S.E_2 = \frac{\sigma}{30} = \frac{\left(\frac{\sigma}{20}\right)}{\left(\frac{3}{2}\right)} = \frac{S.E_1}{\left(\frac{3}{2}\right)}$$

Thus if the sample size is increased from 400 to 900, the standard error will be divided by $\frac{3}{2}$

Problems on central limit theorem:

1. The mean height of students in a college is 155cms and standard deviation is 15. what is the probability that the mean height of 36 students is less than 157 cms.

Solution: Given mean of the population (μ) = 155 cm

Standard deviation of the population (σ) = 15 cm

Sample size (n) = 36

Mean of sample ($\bar{x}$) = 157 cm

Now $Z = \dfrac{\bar{x} - \mu}{\dfrac{\sigma}{\sqrt{n}}}$

$= \dfrac{157 - 155}{\dfrac{15}{\sqrt{36}}}$

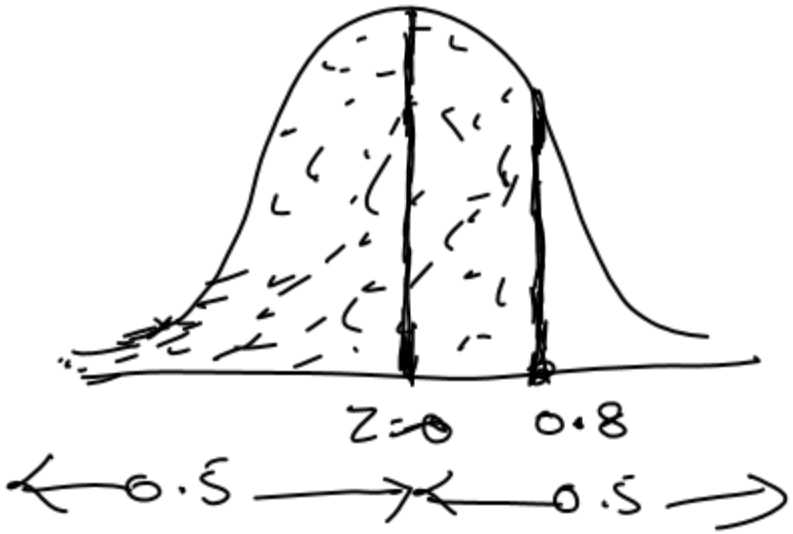
$=0.8$

$$\therefore P(\bar{x} \leq 157) = P(Z < 0.8)$$

$$= 0.5 + P(0 \leq Z \leq 0.8)$$

$$= 0.5 + 0.2881$$

$$\therefore P(\bar{x} \leq 157) = 0.7881$$



2. If a 1-gallon can of paint covers on an average 513 square feet with a standard deviation of 31.5 square feet, what is the probability that the mean area covered by a sample of 40 of these 1-gallon cans will be anywhere from 510to 520 square feet?

Solution :-

$\mu = 513$ sq. feet

$\sigma = 31.5$ sq. feet.

$n = 40$

$P(510 \leq \bar{x} \leq 520) = ?$

If $\bar{x} = 510$, $z = \dfrac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$

$= \dfrac{510 - 513}{\left(\frac{31.5}{\sqrt{40}}\right)} = -0.6$

$$\bar{x} = 520, \quad z = \frac{520 - 513}{\left(\frac{31.5}{\sqrt{40}}\right)} = 1.4$$

$$P(510 \le \bar{x} \le 520) = P(-0.6 \le z \le 1.4)$$

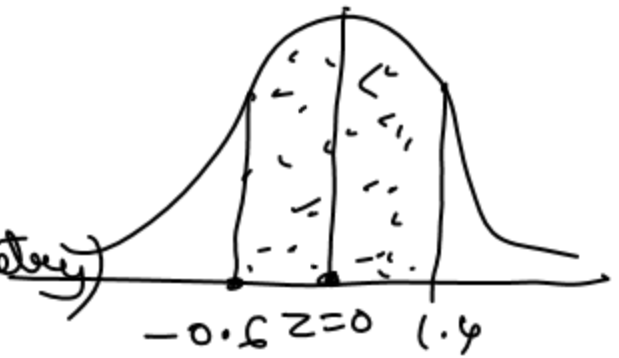
$$= P(-0.6 \le z \le 0) + P(0 \le z \le 1.4)$$

$$= P(0 \le z \le 0.6) + P(0 \le z \le 1.4) \quad \because \text{due to symmetry}$$

$$= 0.2258 + 0.4192$$

$$= 0.645$$

$\therefore P(510 \le \bar{x} \le 520) = 0.645$.



3. The mean of certain normal population is equal to the standard error of the mean of the samples of 64 from that distribution. Find the probability that the mean of the sample size 36 will be negative.

Solution :- mean of population $= \mu$

S.E of sample mean $\bar{x} = \frac{\sigma}{\sqrt{n}}$ where $n = 64$.

Given that

mean of normal population = S.E of $\bar{x}$

$$\mu = \frac{\sigma}{\sqrt{n}} \Rightarrow \mu = \frac{\sigma}{\sqrt{64}} \qquad P(\bar{x} < 0)$$

$$\therefore \mu = \frac{\sigma}{8}$$

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} \quad \text{where } n = 36$$

$$\Rightarrow z = \frac{\bar{x} - \frac{\sigma}{8}}{\frac{\sigma}{\sqrt{36}}}$$

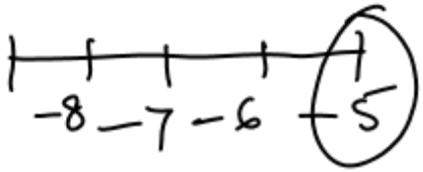
$$z = \frac{\bar{x} - \frac{\sigma}{8}}{\frac{\sigma}{6}} = \frac{\bar{x}}{\left(\frac{\sigma}{6}\right)} - \frac{\left(\frac{\sigma}{8}\right)}{\left(\frac{\sigma}{6}\right)}$$

$P(\bar{x} < 0)$

$z = \bar{x}$ (5)

$\Rightarrow$

$$= \frac{6\bar{x}}{\sigma} - \frac{\sigma}{8} \times \frac{6^3}{4}$$



$$z = \frac{6\bar{x}}{\sigma} - \frac{3}{4}$$

If $\bar{x} < 0$, $z < -5$

$$z = \frac{6\bar{x}}{\sigma} - 0.75$$

If $\bar{x} < 0$ then clearly z will be less than 0.75

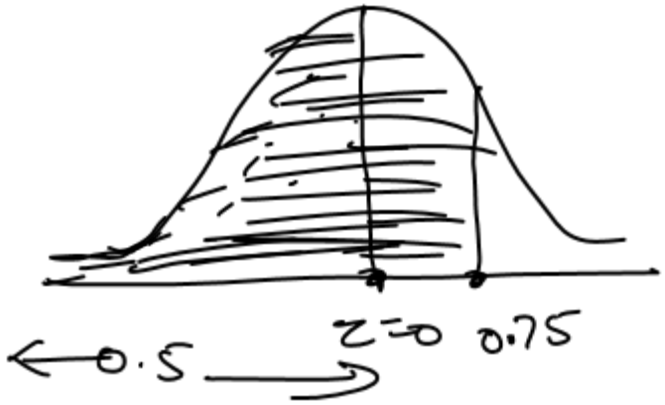
(ii) If $z < 0.75$ then only $\bar{x}$ will be negative.

$$P(\bar{x} < 0) = P(z < 0.75)$$

$$= 0.5 + P(0 < z < 0.75)$$

$$= 0.5 + 0.2734$$

$$P(\bar{x} < 0) = 0.7734.$$



4. A normal population has a mean of 0.1 and standard deviation of 2.1 . Find the probability that the mean of a sample of size 900 will be negative.

Solution:

population mean $\mu = 0.1$

Population standard deviation $= 2.1$

mean of sample $= \bar{x}$

sample size $n = 900$

we have

$$z = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$$

$$z = \frac{\bar{x} - 0.1}{\left(\frac{2.1}{\sqrt{900}}\right)} = \frac{\bar{x} - 0.1}{0.07}$$

$$z = \frac{\bar{x} - 0.1}{0.07}$$

$\bar{x} - 0.1 = 0.07Z$

$\bar{x} = 0.07Z + 0.1$

probability that the sample mean $\bar{x}$ will be negative

$= P(\bar{x} < 0)$

$= P(0.07Z + 0.1 < 0)$

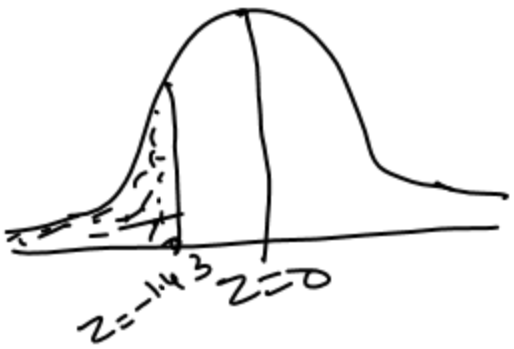
$= P(0.07Z < -0.1)$

$= P\left(Z < \frac{-0.1}{0.07}\right) = P(Z < -1.43)$

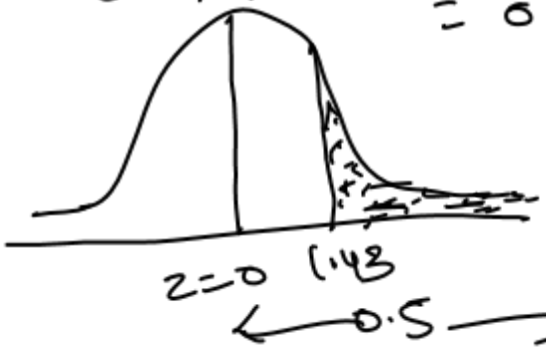
$= 0.5 - P(0 \le Z \le 1.43)$

$= 0.5 - 0.4236$

$= 0.0764$



$\simeq$



$\therefore P(\bar{x} < 0) = 0.0764$

5. A random sample of size 64 is taken from a normal population with mean51.4 and S.D 68. what is the probability that the mean of sample will

a) exceed 52.9 b) fall between 50.5 and 52.3 c) less than 50.6

6. If the mean breaking strength of copper wire is 575lbs, with a standard deviation of 8.3lbs. How large a sample must be used in order that there will be one chance in 100 that the mean breaking strength of the sample is less than 572lbs.

7)The mean voltage of a battery is 15 and S.D is 0.2. Find the probability that four such batteries connected in series will have a combined voltage of 60.8 or more volts.

Solution:

Let the mean voltage of batteries A, B, C, D be $\overline{X_A}, \overline{X_B}, \overline{X_C}, \overline{X_D}$

$$\mu_{\overline{X_A}+\overline{X_B}+\overline{X_C}+\overline{X_D}} = \mu_{\overline{X_A}} + \mu_{\overline{X_B}} + \mu_{\overline{X_C}} + \mu_{\overline{X_D}}$$
$$= 15 + 15 + 15 + 15 = 60$$

$$\sigma_{A+B+C+D} = \sqrt{\sigma_A^2 + \sigma_B^2 + \sigma_C^2 + \sigma_D^2}$$

$$= \sqrt{(0.2)^2 + (0.2)^2 + (0.2)^2 + (0.2)^2}$$
$$= \sqrt{4 \times (0.2)^2} = 0.4$$

Let X be the combined voltage of the series

$$when\ X = 60.8, z = \frac{X - \mu}{\sigma} = \frac{60.8 - 60}{0.4} = 0.2$$

$$p(X \geq 60.8) = p(z > 2) = 0.5 - 0.4772 = 0.0228$$

$= 0.5 - p(0 \leq z \leq 2)$

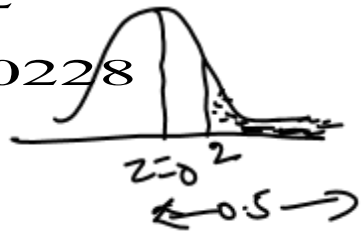


Table : Area under standard normal curve from 0 to $\frac{x-\mu}{\sigma}$

$\frac{x-\mu}{\sigma}$	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	.0000	.0040	.0080	.0120	.0159	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1084	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1916	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2258	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2518	.2549
0.7	.2580	.2611	.2642	.2671	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4430	.4441
1.6	.4452	.4463	.4474	.4485	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4654	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4762	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4865	.4868	.4871	.4875	.5878	.4881	.4884	.4887	.4992
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946.	4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4980	.4980	.4981
2.9	.4981	.4982	.4983	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.1	.4990	.4991	.4991	.4991	.4992	.4992	.4992	.4992	.4993	.4993

Estimation

Estimate:

An estimate is a statement made to find unknown population parameter

Estimator:

The method or rule to determine an unknown population parameter is called an **Estimator.** For example sample mean is an estimator of population mean because sample mean is a method of determining the population mean. A parameter can have many or 1,2 estimators. The estimators should be found so that they are very near to parameter values.

Statistical estimation:

The process by which we draw a conclusion about some measure of a population based on a sample value. The measures might be a variable , such as mean, S.D etc.

Types of estimation:

There are two types of estimates to determine the parameters of the population

1. Point estimation
2. Interval estimation

Point estimation:

An estimate of a population parameter given by a single number is called a **point estimate** of the parameter. If we say that a distance in 5.28 mts , we are giving a point estimate.

Interval estimation:

An estimate of a population parameters given by two numbers between which the parameter may be considered to lie is called an **interval estimate** of the parameter. The distance lie between 5.25 and 5.31 mts.

Unbiased estimator:

A statistic or point estimator $\hat{\theta}$ is said to be an unbiased estimator of the parameter θ if the mean of $\hat{\theta}$ is equal to the population parameter θ i.e.,

$$E(\hat{\theta}) = \theta$$

Properties of good estimator:

The estimator should be

a)consistent

b)efficient

c) sufficient

Maximum error:

$$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \rightarrow \text{large samples}$$

$$E = t_{\alpha/2} \cdot \frac{S}{\sqrt{n}} \rightarrow \text{small samples}$$

$$E = z_{\alpha/2}\sqrt{\frac{PQ}{n}} \quad (\sigma \text{ unknown})$$

$$E = \frac{z_{\alpha/2}}{2\sqrt{n}} \quad [\text{Both } \sigma \text{ \& } P \text{ are unknown}]$$

σ-population S.D

$z_{\alpha/2}$ = value of z leaving an area α/2 to the right

n=sample size

P=population proportion

Q=1-P

E=maximum error

- Confidence interval for μ (for large samples $n \geq 30$) σ known

$$\bar{x} - Z_{\alpha/2}\frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$$

Confidence Interval for Large sample:

$$\left(\bar{x} - z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\cdot\frac{\sigma}{\sqrt{n}}\right) \text{ or } (\bar{x} - E, \bar{x} + E)$$

Confidence Interval for small sample:

$$\left(\bar{x} - t_{\alpha/2}\cdot\frac{s}{\sqrt{n}},\ \bar{x} + t_{\alpha/2}\frac{s}{\sqrt{n}}\right) \text{ or } \bar{x} - E, \bar{x} + E)$$

Confidence percentage: (1- α)100%

Value of $z_{\alpha/2}$:

for 99% , $z_{\alpha/2} = 2.58$

for 98% , $z_{\alpha/2} = 2.33$

For 95%, $z_{\alpha/2} = 1.96$

For 90% $z_{\alpha/2} = 1.645$

A random sample of 400 items is found to have mean 82 and standard deviation of 18. Find the maximum error of estimation at 95% confidence interval. Find the confidence limits for the mean if $\bar{x}$ =82.

Solution:- Sample size $n = 400$

Sample mean $\bar{x} = 82$

Standard deviation $s = 18$

Maximum error $(E) = ?$

At 95% confidence $z_{\alpha/2} = 1.96$

$$E = z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = 1.96 \times \frac{18}{\sqrt{400}}$$

$$E = 1.76$$

95% confidence interval $= \left(\bar{x} - z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right)$ or $(\bar{x} - E, \bar{x} + E)$

$$= \left(82 - 1.96 \times \frac{18}{\sqrt{400}} \;,\; 82 + 1.96 \times \frac{18}{\sqrt{400}}\right)$$

$$= (80.24,\ 83.76)$$

$$\text{or } \left(\bar{x} - E,\ \bar{x} + E\right)$$

$$= (82 - 1.76,\ 82 + 1.76)$$

$$= (80.24,\ 83.76)$$

5. Determine a 95% confidence interval for the mean of normal distribution with variance 0.25, using a sample of size 100 values with mean 212.3.

Solution: Given Sample size (n) = 100 variance $\sigma^2 = 0.25$

$\therefore$ Standard deviation of sample (σ) = $\sqrt{0.25} = 0.5$

Mean of sample ($\bar{x}$) = 212.3 and $Z_{\alpha/2} = 1.96$(for 95%)

$\therefore$ 95% Confidence interval = $\left(\bar{x} - Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}, \bar{x} + Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \right)$

$$= \left(212.3 - 1.96 \cdot \frac{0.5}{\sqrt{100}}, 212.3 + 1.96 \cdot \frac{0.5}{\sqrt{100}} \right)$$

$$= (212.202, 212.398)$$

3. What is the size of the smallest sample required to estimate an unknown proportion to within a maximum error of 0.06 with at least 95% confidence.

Solu: sample size $n = ?$

Maximum error $E = 0.06$

For 95% confidence, $z_{\alpha/2} = 1.96$

$$E = \frac{z_{\alpha/2}}{2\sqrt{n}}$$

$$\Rightarrow \sqrt{n} = \frac{z_{\alpha/2}}{2E}$$

$$n = \left(\frac{z_{\alpha/2}}{2E}\right)^2$$

$$= \left(\frac{1.96}{2 \times 0.06} \right)^2$$

$$n = 266.78 \approx 267$$

$\therefore$ Sample size $= 267$.

4. In a study of an automobile insurance a random sample of 80 body repair costs had a mean of Rs. 472.36 and the standard deviation of Rs. 62.35. If $\bar{x}$ is used as a point estimate to the true average repair costs, with what confidence we can assert that the maximum error doesn't exceed Rs. 10.

Solution: Given Sample size (n) = 80

Standard deviation of sample (s) = 62.35

Mean of sample ($\bar{x}$) = 472.36

Maximum Error(E)=10

$$\therefore E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow Z_{\alpha/2} = \frac{E.\sqrt{n}}{\sigma} = \frac{10\sqrt{80}}{62.35} = 1.4345$$

The area when z=1.43 from the tables is 0.4536

$$\therefore \frac{\alpha}{2} = 0.4236 \Rightarrow \alpha = 0.8472$$

$$\therefore \text{Confidence} = (1-\alpha)100\ \% = 84.72\ \%$$

Hence we are 84.72% confidence that the maximum error is Rs. 10.

5.A random sample of size 100 has a standard deviation of 5. what can you say about the maximum error with 95%confidence.

6. The mean and standard deviation of population are 11,795 and 14,054 respectively. If n=50, find the 95% confidence interval for the mean.

7.What is the maximum error one can expect to make with the probability 0.90 when using the mean of a random sample of size n=64 to estimate the mean of the population with variance 2.56.

Solution:

Sample size n=64

Population variance $\sigma^2 = 2.56$

Standard deviation $\sigma = \sqrt{2.56} = 1.6$

$For\ .90\ probability\ \ z_{\alpha/2} = 1.645$

$$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

8. It is desired to estimate the mean number of hours of continuous use until a certain computer will first require repairs. If it can be assumed σ=48 hours, how large a sample be needed so that one will be able to assert with 90% confidence that the sample mean is off by at most 10 hours.

Solution: $\sigma = 48$

For 90%, $Z_{\alpha/2} = 1.645$

$n = ?$

$E = 10$ hours

$$E = Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}} \Rightarrow \sqrt{n} = \frac{Z_{\alpha/2} \cdot \sigma}{E}$$

$$n = \left(\frac{Z_{\alpha/2} \cdot \sigma}{E}\right)^2 = \left(\frac{1.645 \times 48}{10}\right)^2 = 62.34 \simeq 62$$

9. Find 95% confidence limits for the mean of normality distributed population from which the following sample was taken 15,17,10,18,16,9,7,11,13,14.

Soln:- $\left(\bar{x} - z_{\alpha/2}\frac{S}{\sqrt{n}}, \bar{x} + z_{\alpha/2}\frac{S}{\sqrt{n}}\right)$ → Confidence limits

95% $z_{\alpha/2} = 1.96$

$n = 10$, $\bar{x} = \frac{15+17+10+18+16+9+7+11+13+14}{10}$ $\quad \frac{\sum x_i}{n}$

$\therefore \bar{x} = 13$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

$$= \frac{(15-13)^2 + (17-13)^2 + (10-13)^2 + (18-13)^2 + (16-13)^2 + (9-13)^2 + (7-13)^2 + (11-13)^2 + (13-13)^2 + (14-13)^2}{9}$$

$s^2 = \frac{40}{3}$ $\quad \therefore S = 3.65$

95% Confidence limits $= \left(\bar{x} - z_{\alpha/2}\cdot\frac{S}{\sqrt{n}},\ \bar{x} + z_{\alpha/2}\cdot\frac{S}{\sqrt{n}}\right)$

$= \left(13 - 1.96 \times \frac{3.65}{\sqrt{10}},\ 13 + 1.96 \times \frac{3.65}{\sqrt{10}}\right)$

$= (10.74, 15.26)$

(10) If we can assert with 95% that the maximum error is 0.05 and $P = 0.2$, find the size of the sample.

Solution: $P = 0.2 \quad Q = 1 - P = 1 - 0.2 = 0.8$

For 95%, $z_{\alpha/2} = 1.96$,

Maximum Error $E = 0.05$

$$E = z_{\alpha/2}\cdot\sqrt{\frac{PQ}{n}} \Rightarrow \sqrt{n} = z_{\alpha/2}\frac{\sqrt{PQ}}{E}$$

$$n = \left(\frac{z_{\alpha/2}}{E}\right)^2 \times PQ = \left(\frac{1.96}{0.05}\right)^2 \times 0.2 \times 0.8 = 246.$$

$\therefore$ Sample size $= 246$